GenScale: A Benchmark for Relative Object Scale in Image Generation and Editing

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https://github.com/lingxiao-li/GenScale

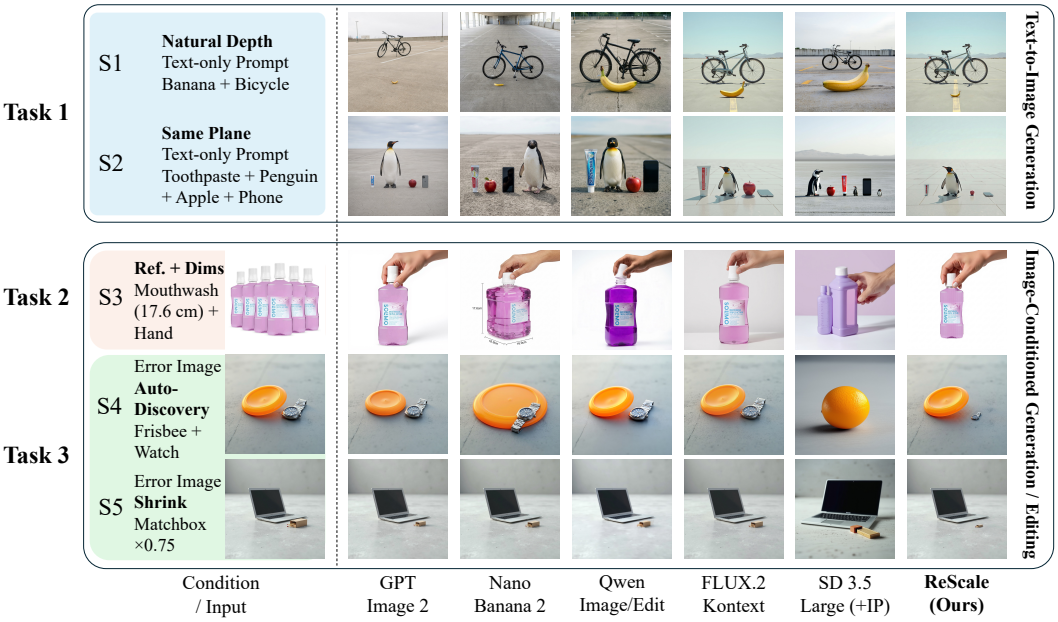

Figure 1: **Relative object scale remains a challenge for modern image generators.** S1–S5: text-only common-object generation under natural depths or same-plane layouts, human-product generation from a product reference with metric dimensions, and scale-error correction from erroneous source images with either auto-discovery or a precise resize instruction. The general-purpose image generators/editors achieve high visual realism and yet often violate real-world size relationships, whereas ***Rescale*** produces more plausible relative scale. Zoom in for the best viewing.

Abstract

Modern image generation and editing systems can produce photorealistic, prompt-aligned images, but still often render familiar objects at implausible relative sizes. To measure this failure mode, we introduce ***GenScale***, a benchmark and evaluation protocol for real-world relative object scale in image generation and editing. *GenScale* contains 900 image-level entries and 1,643 pairwise anchor-target scale relations across common-object generation, human-product generation with metric dimensions, and scale correction from failed generations. We further design a human-calibrated ordinal judge for scalable pairwise scale evaluation. Last but not the least, we introduce ***Rescale***, a model-agnostic post-processing agent for localized scale correction without modifying the source generator. Experiments reveal that state-of-the-art image generators and editors cannot reliably observe relative scale yet, while *Rescale* consistently improves scale plausibility across generated and edited images. Together, *GenScale* establishes relative object scale as a distinct, measurable, and actionable capability for image generation systems.

Preprint.

1 Introduction

Recent text-to-image (T2I) generation has advanced rapidly, with diffusion-based models [53], autoregressive generators [72], and transformer-based systems [11] achieving strong visual fidelity, stylistic diversity, and prompt adherence. However, these advances do not guarantee a basic requirement of physical realism: rendering objects at plausible *relative scale*. An image may look photorealistic and prompt-aligned while still placing familiar objects at mutually implausible sizes. This issue is increasingly important as generated images are used in advertising, product visualization, virtual character creation, and professional content production, where incorrect object proportions can immediately undermine visual credibility.

As illustrated in Fig. 1, the failure appears across text-only generation, reference-conditioned human–product generation, and scale correction, where models often preserve identity and visual realism while violating real-world relative scale. Fig. 2 further shows that these errors are systematic rather than anecdotal: small real-world objects are often rendered too large, while large objects are rendered too small. We refer to this recurring pattern as *mean regression in generated object scale*, a failure mode that affects visual realism, product visualization, synthetic data construction, and scale-aware editing.

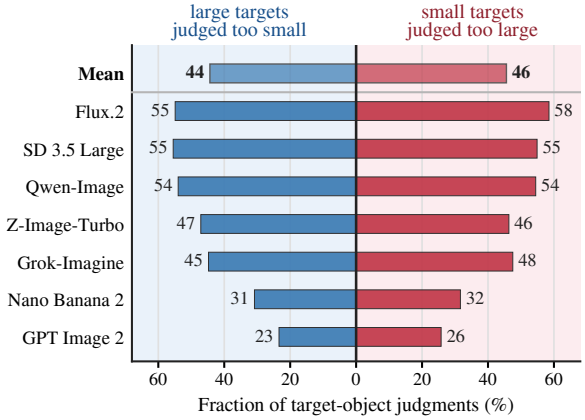

Figure 2: **Mean-regression errors on *GenScale* Task 1.** Bars show two directional scale failures (400 tests) across generators: large real-world target objects are rendered too small, while small target objects are rendered too large. The consistent pattern indicates that current generators tend to compress object scale toward a typical visual size rather than preserving real-world size ratios.

Existing benchmarks evaluate related capabilities, including object presence, counting, and attribute binding [15], compositional prompt following [27], commonsense plausibility [13], physical reasoning [39], and spatial relation control [64]. Yet scale is usually subsumed under broader semantic, physical, or spatial reasoning rather than evaluated as a distinct object-level relation. The finer-grained question we study is *relative object scale*: when several objects appear in a single image, **can a generative model render their size relationships in a way that faithfully reflects the real world?** This question differs from general physical plausibility because a scene may satisfy coarse commonsense and layout constraints while still assigning an implausible size ratio to two objects.

This motivates ***GenScale***, a benchmark for real-world relative object scale in image generation and editing. *GenScale* represents scale as a pairwise anchor-target relation grounded in external object-size metadata, and covers three regimes: implicit common-object size priors, human-product metric scale, and post-generation scale correction. It contains 900 image-level entries and 1,643 anchor-target relations, enabling dense pairwise evaluation while retaining image-level grouping for model comparison. Tab. 1 summarizes its coverage relative to representative prior benchmarks.

Evaluating these anchor-target relations requires more than measuring pixel-space ratios, since scale depends jointly on object identity, physical size, perspective, depth, occlusion, and placement. *GenScale* therefore uses a human-calibrated ordinal evaluation protocol rather than a raw pixel-ratio metric. We calibrate a Gemini-based judge against annotations from nine human raters, achieving strong agreement with human consensus.

We further introduce ***Rescale***, a model-agnostic size-correction agent that uses structured scale information to execute localized, geometry-aware resizing and reinsertion. Ordinary users can use it to post-correct relative scale issues of any generated or edited images, and model developers may use it to enrich their training data (e.g., identifying hard examples using our calibrated judge, fixing their relative scale using *Rescale*, and mixing them back to the training set).

In the experiments, we benchmark state-of-the-art image generators and editors, revealing persistent relative-scale failures and showing that *Rescale* consistently improves scale plausibility across generated and edited images.

In summary, our contributions are four-fold:

Table 1: **Comparison with representative related benchmarks.** Prior benchmarks cover individual axes such as composition, physical plausibility, or spatial reasoning, whereas *GenScale* explicitly unifies object-object relative scale, human-product metric size, and post-generation scale correction.

	Broad Capabilities			**Scale-focused Evaluation**		
Work	**Compos.**	**Physical**	**Spatial**	**Obj.-obj. scale**	**Human-prod. size**	**Correction**
GenEval [15]	✓	✗	✓	✗	✗	✗
T2I-CompBench++ [27]	✓	✗	✓	✗	✗	✗
GenAI-Bench [34]	✓	✗	✗	✗	✗	✗
PhyBench [39]	✗	✓	✗	✗	✗	✗
GenSpace [64]	✗	✗	✓	✓	✗	✗
GenScale	✓	✓	✓	✓	✓	✓

- We introduce ***GenScale***, the first benchmark dedicated to relative object scale in image generation and editing, covering common-object scale priors, human-product metric scale, and post-generation scale correction.
- We develop a **human-calibrated ordinal evaluation protocol** for scalable pairwise scale judgment, enabling reliable assessment of whether generated object-object proportions match real-world size relationships.
- We propose ***Rescale***, a model-agnostic scale-correction agent that uses structured scale metadata to perform localized, geometry-aware editing without modifying the source generator.
- We benchmark state-of-the-art image generators and editors, showing that they cannot reliably observe relative scale, while *Rescale* consistently improves scale plausibility across the generated and edited images.

2 Related Work

2.1 Image Generative Models.

Text-to-image generation has progressed from diffusion and autoregressive models [23, 58, 51, 54, 53, 72] to transformer-based diffusion and rectified-flow systems [46, 11], substantially improving visual fidelity, prompt adherence, and scalability. Recent open and closed models, including FLUX.2 [4], Qwen-Image [66, 47], Z-Image [60], Seedream 4.5 [6], GPT Image 2 [43], and Nano Banana 2 [16], further extend these capabilities to high-fidelity generation, reference conditioning, and image editing. This progress motivates evaluating relative scale as a distinct capability rather than assuming it follows from overall visual quality.

2.2 Evaluation Benchmarks for Text-to-image Generation.

Early evaluation relied on distributional quality metrics such as FID and IS [22, 55], image-text similarity metrics such as CLIPScore [21], and curated prompt suites such as DrawBench and PartiPrompts [54, 72]. More targeted benchmarks evaluate compositional generalization [45], object presence, counting, color, position, and attribute binding [15], open-world composition and numeracy [26, 27], complex prompt following [34], reasoning [7], long-prompt control [31], factuality [28], and world-knowledge-informed semantic evaluation [42]. Automatic evaluation increasingly uses VLM-based protocols [25, 37, 34], but recent analyses show that off-the-shelf LMM judges require task-specific validation for nuanced generated-image judgments [75]. In contrast, *GenScale* uses a human-calibrated ordinal protocol for scale-specific pairwise judgment.

2.3 Spatial, Physical, and Commonsense Evaluation.

Relative object scale is related to spatial reasoning and physical commonsense, but existing benchmarks usually treat it as part of broader capabilities. Commonsense-T2I [13] evaluates everyday commonsense consistency, PhyBench [39] targets physical commonsense errors, and Generate Any Scene [14] evaluates structured object and relation control through scene graphs. Most closely related to our work, GenSpace [64] benchmarks spatially aware image generation through pose, spatial-relation, and metric-measurement tasks, while recent spatial-intelligence benchmarks emphasize object arrangement and layout consistency [65]. *GenScale* complements these benchmarks by isolating real-world relative object scale as a pairwise, physically grounded evaluation target, with additional coverage of human-product metric scale and post-generation scale correction.

3 GenScale: A Benchmark for Physically Grounded Relative Scale

GenScale evaluates physically grounded relative scale as a pairwise anchor-target relation with structured metadata, including object identities, physical reference lengths, expected 3D scale ratios,

Table 2: **Overview of *GenScale*.** *GenScale* evaluates physically grounded relative scale across implicit common-object priors (T1), explicit metric product scale (T2), and scale correction (T3). Image-level entries may contain multiple anchor-target pairs.

Task	Scenario	Images	Pairs	Capability tested
T1	S1: Natural Depth	200	570	Implicit scale priors under perspective
T1	S2: Same Plane	200	573	Implicit scale priors with depth controlled
T2	S3: Human-Product	300	300	Metric product scale with human anchors
T3	S4: Auto-Discovery	100	100	Diagnose scale error and choose resize
T3	S5: Precise Instruction	100	100	Execute exact numeric resize factor
Total		900	1,643	Generation, customization, and correction

scenario labels, and product reference images when applicable. It contains 900 image-level entries and 1,643 pairwise scale relations across three tasks and five scenarios: common-object generation (Task 1 or **T1**), human-product scale realization (**T2**), and post-generation scale correction (**T3**). Each anchor-target pair is used as the atomic evaluation unit. Tab. 2 summarizes the benchmark taxonomy, and Fig. 3 illustrates the construction pipeline and input conditions detailed in the following.

3.1 Scale Metadata

GenScale is grounded in two types of physical-size metadata. For common-object scale reasoning (T1), we construct a category-level knowledge base from COCO [36] and LVIS [20], retaining visually identifiable categories with relatively stable physical extent and removing classes with large intra-class variation, ambiguous semantics, or non-rigid size, such as *person*, *dog*, *bird*, *tree*, *bag*, *chair*, and *boat*. This pool contains 97 non-human common objects and is used for Task 1 and the common-object correction cases in Task 3. For human-product scale (T2), we use product-level dimensions and reference images from Amazon Berkeley Objects (ABO) [9], together with human anchors including hand, head/face, foot/leg, and full body. Object and anchor dimensions are grounded in cross-domain sources, including anthropometric surveys [19], biological and food references [41, 61], standardized manufactured-object specifications [30, 29, 1], and traffic and sport-object specifications [12, 38]. For each entry, we store a characteristic length, an acceptable physical range when available, and the source used to ground the measurement. Check Sec. C for more details.

3.2 Benchmark Construction

As shown in Fig. 3, the input to image generators differs by tasks: An input in Task 1 consists of only a text prompt, Task 2 uses a product reference image plus a text prompt, and Task 3 pairs an erroneous source image with one of two edit prompts.

Task 1: Implicit Common-Object Scale. Task 1 is text-only image generation that tests whether models can infer relative sizes of familiar objects without explicit metric cues. Each image-level entry contains a prompt with two to four non-human objects sampled from the common-object pool; no object-specific numeric dimensions or reference images are given to the model. We retain object sets whose largest-to-smallest characteristic-length ratio lies in $[4, 20]$, avoiding near-equal comparisons that are hard to judge and extreme ratios where layout constraints dominate. The generated image is then evaluated through all valid anchor-target pairs induced by the objects in the prompt.

Task 1 has two scenarios. In **S1: Natural Depth**, objects may appear with realistic perspective, mild occlusion, and depth ordering, reflecting ordinary generation settings where scale errors can be hidden by near-far placement. In **S2: Same Plane**, objects are constrained to approximately the same depth plane, reducing perspective ambiguity and making image-space proportions more directly reflect real-world scale ratios.

Task 2: Explicit Metric Scale in Human-Product Interaction. Task 2 is an image-conditioned human-product generation task. Each entry contains a product reference image and a prompt specifying the product's metric dimensions and a size-conditioned human anchor, requiring the model to preserve product identity while rendering plausible scale relative to the human body. This tests product-level scale realization, where category priors are insufficient because visually similar commercial products can have different dimensions. Products shorter than 30 cm are paired with a hand, products between 30 and 60 cm with a head/face anchor, products between 60 and 100 cm with a foot/leg anchor, and products larger than 100 cm with a full-body anchor. Prompts require natural, use-consistent interaction and catalog-style visibility, so generators must translate metric dimensions into plausible human-object proportions rather than merely copy the reference appearance.

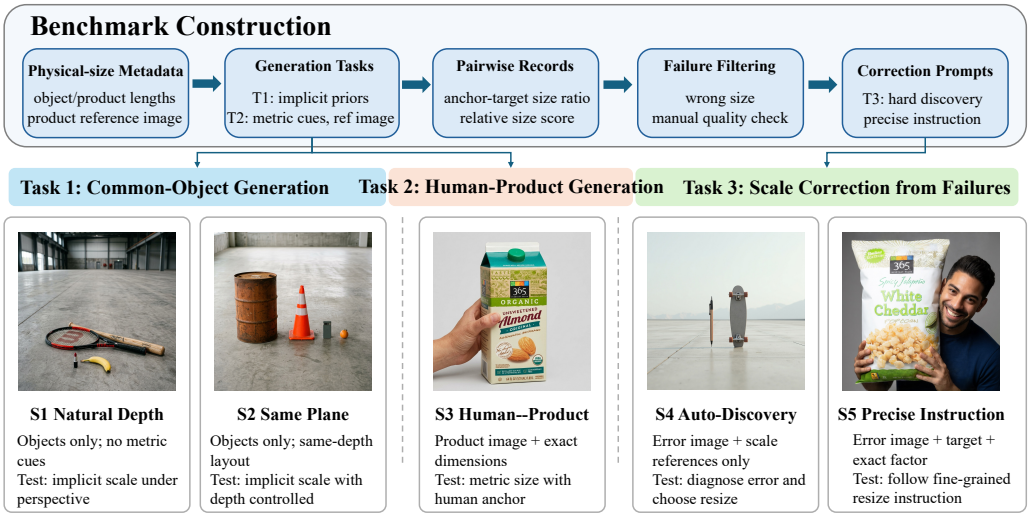

Figure 3: ***GenScale* benchmark construction.** *GenScale* is built from physical-size metadata and product references, then instantiated as three tasks covering common-object generation, human-product metric scale, and scale correction. **Task 3** converts non-plausible but editable outputs from **Tasks 1** and **2** into S4 Auto-Discovery and S5 Precise Instruction, which separately test automatic scale-error diagnosis and exact numeric resize following.

Task 3: Scale Correction from Failed Generations. Task 3 converts failed generations from Tasks 1 and 2 into scale-aware image editing tasks. Each source case contains an erroneous image and one pairwise relation whose relative scale is judged implausible under the pairwise criterion introduced in Sec. 3.3. We use FLUX.2 [4] generations for common-object scenes and Qwen-Image [66] generations for human-product scenes, retaining only pairwise cases: two-object Task 1 images and one product-human relation in Task 2. The corresponding pairwise record provides the rendered object-size ratio and target physical ratio, from which we derive the fixed reference object, editable object, resize direction, and multiplicative scale factor. We remove images with missing or unidentifiable objects, duplicated or merged objects, severe artifacts, heavy occlusion or blur, and extreme correction factors that make localized editing ill-defined; this filtering only ensures that each retained source image contains a scorable and editable scale error.

Each of the 100 retained source images is expanded into two benchmark entries with the same erroneous image but different edit prompts. In **S4: Hard Auto-Discovery**, the model receives the erroneous image and only the information needed to judge scale, but not the editable object, resize direction, or scale factor; it must diagnose the error and choose a local correction. For common-object sources, this includes deciding which object should change, while for human-product sources the product is editable and the human body part is fixed. In **S5: Precise Scale Instruction**, the prompt directly gives the editable object, reference object, resize direction, and exact scale factor, isolating fine-grained numeric resize following from visual error diagnosis.

3.3 Human-Calibrated Scale Evaluation

GenScale evaluates relative scale at the object-pair level. A raw object-space ratio is insufficient because the same apparent ratio may be plausible or implausible depending on object identity, depth ordering, foreshortening, occlusion, and camera perspective. We therefore develop a human-calibrated ordinal protocol: Each generated image is decomposed into anchor–target pairs from the *GenScale* metadata; the anchor is treated as the reference, and the evaluator judges whether the target has a plausible real-world size relative to it.

Pairwise Ordinal Rubric. For each anchor-target pair, evaluators are shown the image, object names, and reference physical lengths. They assign a five-point ordinal score: 1/2 indicate that the target is severely/slightly undersized, 3 indicates physically plausible scale, and 4/5 indicate that it is slightly/severely oversized. Operationally, score 3 corresponds to an estimated scale error within approximately $\pm 20\%$, scores 2/4 to errors between 20% and 60%, and scores 1/5 to errors larger than 60%. Evaluators are instructed to account for perspective, occlusion, partial visibility, and foreshortening. A pair is marked invalid only when reliable scale judgment is impossible, e.g., because an object is missing, merged, ambiguously duplicated, or too degraded to identify.

Table 3: **Human Reliability and Gemini-Human Alignment.** Exact and ≤ 1 denote agreement on the five-point ordinal scale defined as exact match and difference by at most 1, respectively. MAE is the mean absolute ordinal deviation. r is Pearson correlation on 1-5 scores. QWK denotes quadratic-weighted kappa. Human rows report mean$\pm$std across nine annotators.

Evaluator / Reference	Split	Exact ↑	≤ 1 ↑	MAE ↓	r ↑	QWK ↑
Human vs. Aggregate Consensus	T1+T2	65.1±7.0	94.0±3.2	0.413±0.099	0.764±0.076	0.754±0.077
Human vs. LOO Consensus	T1+T2	58.0±6.6	93.3±3.0	0.491±0.093	0.719±0.069	0.706±0.070
Gemini vs. Human Consensus	T1+T2	63.95	97.15	0.389	0.833	0.823
Gemini vs. Human Consensus	T1	61.83	96.49	0.417	0.847	0.837
Gemini vs. Human Consensus	T2	73.00	100.00	0.270	0.470	0.468

Human Consensus. We collect annotations from nine human raters on a calibration split sampled from Tasks 1 and 2, covering natural-depth common-object scenes, same-plane common-object scenes, and human-product scenes. After aligning completed annotations and retaining visible, scorable pairs, the calibration set contains 285 images and 527 anchor-target pairs. For each pair, we define human consensus as the modal score, with ties broken by the median. As shown in Tab. 3, individual raters agree strongly with the aggregate consensus, and the leave-one-rater-out comparison remains stable, indicating that the consensus is not dominated by any single annotator.

Automatic Judge Calibration. Since exhaustive human evaluation is impractical for large-scale model comparison, we calibrate a Gemini-based VLM judge against the human consensus. The judge receives the same core evidence as human raters—the generated image, anchor and target names, physical reference lengths, the expected 3D length ratio, and the five-point rubric. The generation prompt, when provided, is used only to disambiguate intended objects or layout, not as evidence that the rendered scale is correct. We use scenario-specific judge prompts, with Task 3 inheriting the judge type of its source case, and provide the full prompts in Sec. D. Before scoring scale, we first check whether the required objects are visible, identifiable, and unambiguous, following the invalid-pair rule above; failed pairs are omitted from valid-pair counts and scale metrics, whereas visible but incorrectly scaled pairs are still scored. For each remaining pair, we query the judge five times and aggregate scores by majority vote, using the median when no unique mode exists, which reduces sensitivity to isolated outlier judgments. Tab. 3 shows that the calibrated judge reaches inter-human-level alignment with the consensus, with especially strong within-one agreement and QWK on the full calibration split. Fig. 4 further shows that remaining disagreements are ordinally local rather than directionally reversed. The lower correlation metrics on Task 2 mainly reflect label concentration near the plausible-scale score rather than systematic judge failures. We therefore use the calibrated Gemini judge as the default evaluator for large-scale *GenScale* comparisons and report valid image and pair counts for all systems.

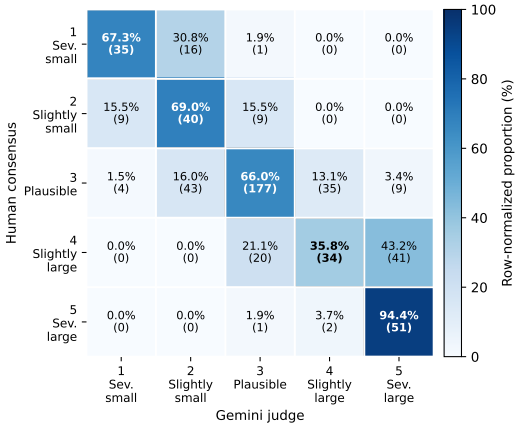

Figure 4: **Disagreement structure of the calibrated Gemini judge.** The row-normalized confusion matrix shows that Gemini-human disagreements are concentrated near the diagonal, with only 2.85% of pairs differing by two or more score levels and no coarse under-to-over or over-to-under sign-flip errors.

4 *Rescale*: Agentic Relative-Scale Correction

The *GenScale* formulation makes relative-scale errors actionable beyond evaluation: We introduce ***Rescale***, a model-agnostic post-processing pipeline that repairs scale inconsistencies in generated images without modifying the source generator. Given an image together with object identities and physical-size references from the prompt or benchmark metadata, *Rescale* changes only the implausibly scaled objects while preserving identity, layout, lighting, and background.

At inference time, a multimodal agent grounds relevant objects and converts pairwise scale evidence into an edit plan. For common-object scenes (T1), it aggregates inconsistent anchor-target relations into a conservative object-level plan and edits over multiple rounds; for human-product scenes (T2), the product is the only editable target while the human body part is fixed. For each edit, the agent

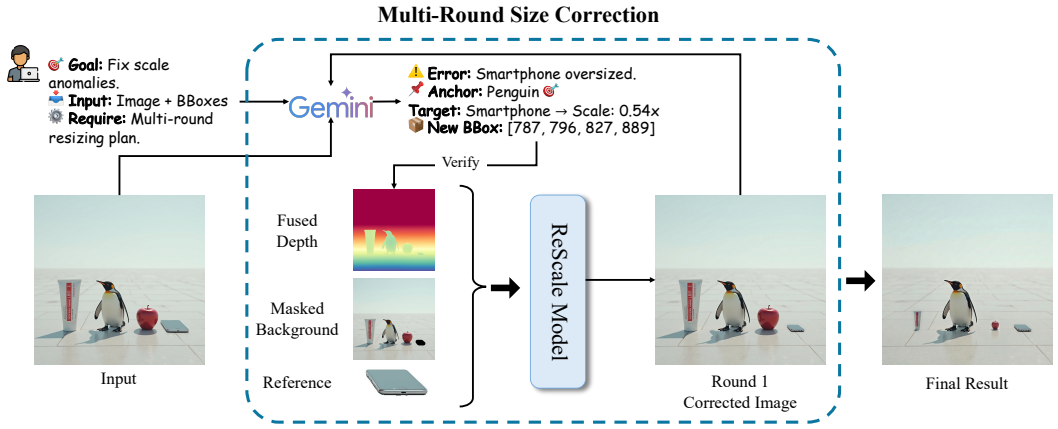

Figure 5: **Agentic multi-round inference in *Rescale*.** Given a generated image and object/scale metadata, the agent diagnoses a scale anomaly, selects the target and anchor, predicts a resize factor and contact-preserving anchor point, and executes localized correction through extraction, background completion, mask/depth adjustment, and modular insertion. The edited result is verified for additional correction rounds when needed.

Table 4: **Task 1: Common-Object Relative Scale.** Results are reported for S1, S2, and both. Err. is scale error, mean $|s-3|$; Plaus. is the percentage of plausible relations (score $s=3$); MR is mean-regression error, where small targets are judged too large, or vice versa.

Model	S1			S2			S1 + S2		
	Err. ↓	Plaus. ↑	MR ↓	Err. ↓	Plaus. ↑	MR ↓	Err. ↓	Plaus. ↑	MR ↓
Nano Banana 2 [16]	0.51	65.1	15.8	0.69	47.8	48.1	**0.60**	**56.5**	31.8
GPT-Image-2 [43]	0.78	51.1	12.2	0.63	52.8	36.0	0.71	51.9	**24.1**
Z-Image-Turbo [60]	0.64	53.2	36.2	0.88	38.1	55.9	0.75	46.1	45.3
Grok Image [68]	0.60	58.8	27.0	0.98	34.7	61.2	0.79	46.7	44.2
Qwen-Image 2512 [47]	0.70	53.3	37.7	1.10	27.2	67.6	0.90	40.0	52.9
FLUX.2 [4]	0.84	44.9	40.1	1.20	23.4	69.5	1.03	33.9	55.1
SD3.5-Large [59]	0.96	36.6	47.0	1.18	24.9	70.0	1.07	31.1	57.8

selects a target o_t and anchor o_a, refines the target box b_t, and estimates a resize factor α by comparing the apparent target-anchor ratio with the plausible real-world ratio under perspective, depth, visibility, contact, boundary, and collision constraints. The plan $\pi = \{o_t, o_a, b_t, \alpha, p\}$ specifies the target, anchor, box, resize factor, and contact-preserving anchor point p; if the multimodal agent detects no reliable scale inconsistency or finds the edit unsafe, *Rescale* returns no edit.

The planned edit is executed by a modularized, local insertion pipeline shown in Figure 5. We segment the target with SAM 2 [52], extract it as an identity reference, remove the original instance to obtain a completed background, construct an enlarged edit mask around the resized box, and estimate monocular depth with DepthAnythingV2 [70]. These steps produce backend-agnostic conditions, including the reference crop, completed background, resized mask, text instruction, and, when supported, fused depth. Our default backend is InsertAnything [57], but it can be replaced by other insertion or inpainting editors. After each edit, the agent verifies whether an obvious scale error remains and applies another round of editing only when needed. Sec. E describes *Rescale* and some ablation studies in detail.

5 Evaluation Results

5.1 Benchmarking State-of-the-Art Models on *GenScale*

We benchmark representative open- and closed-source generative/editing models using the calibrated judge from Sec. 3.3. All metrics are computed on valid pair-level judgments after the visual-quality filter (see Sec. 3.3), and the scale metrics measure performance on those valid judgments: For score $s \in \{1, \ldots, 5\}$, with 3 denoting the plausible scale, *Scale error* is mean $|s-3|$.

Task 1: Common-Object Relative Scale. Tab. 4 shows that the common-object relative scale remains far from solved: even the top-performing model, Nano Banana 2 [50], reaches only 56.5% plausible score. The dominant error is mean regression (from score $s=3$), where small objects are enlarged and large objects are shrunk. The gap between S1 (natural depth) and S2 (same plane)

FLUX.2 Nano Banana 2 GPT Image 2

Size bucket	Objects	Depth rank ↑
Smallest	1,367	0.175
Middle	1,039	0.451
Largest	1,362	0.863
Smaller closer		79.2%
$\rho(\log \ell, \text{depth})$		0.558

Figure 6: **Depth-mediated scale compression in Task 1 S1.** S1 examples show that image generators often place smaller objects closer and larger objects farther away. In the table, we group objects according to real-world size l and then computes their depth ranks in images averaged per group.

Table 5: **Task 2: Human-Anchored Product Scale.** Each prompt contains one product-human scale relation. Scale error is mean $|s-3|$; Plausible denotes score $s=3$; Severe denotes scores of 1 or 5. Full error breakdown, including too-small and too-large rates, is reported in Sec. F

Model	Valid img. / pairs	Scale error ↓	Plausible (%) ↑	Severe (%) ↓
GPT-Image-2 [43]	294 / 294	**0.231**	**78.9**	**2.0**
Nano Banana 2 [16]	295 / 295	0.268	75.6	2.4
Seedream v4.5 [6]	295 / 295	0.302	74.9	5.1
Qwen-Image-Edit-2511 [48]	297 / 297	0.327	71.0	3.7
FLUX.1 Kontext-dev [2]	291 / 291	0.423	63.2	5.5
SD3.5-Large + IP-Adapter [59, 71]	270 / 270	0.600	52.6	12.6

suggests that the natural depth can hide the relative scale errors: models score higher in S1 thanks to perspective and depth ordering of objects of various sizes, whereas S2 exposes object-size errors directly on the same image plane. Fig. 6 supports this interpretation, showing that generators tend to place smaller objects closer and larger objects farther away. Thus, current generators often absorb unrealistic scale ratios through layout choices instead of preserving real-world object-size relations.

Task 2: Human-Anchored Product Scale. Tab. 5 indicates that explicit metric cues and human anchors make scale realization easier, but not solved. Most models achieve higher plausible rates than in Task 1, but the errors are strongly asymmetric: Products are more often enlarged than shrunk. This product-magnification bias can practically hinder the image generators' applications in advertising, virtual reality, and others.

Task 3: Scale-Error Correction. Tab. 6 shows that general-purpose image editors can reduce scale errors on failed generations to very limited degrees. The S4-S5 split reveals a clear diagnosis-execution gap: models perform much better when given the target object, reference object, resize direction, and scale factor, but they are substantially weaker when they must discover the scale error and choose the correction autonomously. Hence, these general-purpose image editors can follow explicit localized resize instructions more reliably than they can infer physically implausible scale.

Overall, *GenScale* exposes three distinct failure modes: mean regression in common-object generation, product magnification in human-product generation, and weak autonomous diagnosis in scale correction. These results support the central claim that relative scale is not subsumed by visual fidelity or prompt adherence, but remains a separate unsolved capability.

5.2 *Rescale* Correction Results

We evaluate *Rescale* on image pairs before and after applying *Rescale* using the calibrated *GenScale* judge. Tabs. 7 and 8 show that *Rescale* reduces scale errors for every Task 1 and Task 2 source

Table 6: **Task 3: Scale-Error Correction.** Results are reported for S4, S5, and both. S4 requires automatic error discovery; S5 provides the target object, reference object, correction direction, and scale factor. Gain is the matched-pair reduction in scale error relative to the before-edit image. B / W denotes the number of pairs with better / worse scale score after editing. Full results in Sec. F.4.

Model	S4			S5			S4 + S5				
	Err. ↓	Plaus. ↑	Gain ↑	Err. ↓	Plaus. ↑	Gain ↑	Valid	Err. ↓	Plaus. ↑	Gain ↑	B / W
Before edit	1.27	20.4	–	1.27	20.4	–	200 / 200	1.27	20.4	–	–
GPT-Image-2	0.96	40.2	+0.33	0.41	66.3	+0.87	195 / 195	**0.68**	**53.3**	**+0.60**	91 / 9
Nano Banana 2	1.03	31.3	+0.24	0.93	33.7	+0.34	197 / 197	0.98	32.5	+0.29	55 / 11
FLUX.1 Kontext-dev	1.29	21.2	-0.02	0.96	36.2	+0.28	193 / 193	1.13	28.5	+0.13	35 / 16
Qwen-Image-Edit-2511	1.25	23.2	+0.02	1.00	34.1	+0.25	190 / 190	1.13	28.4	+0.13	32 / 9
Seedream v4.5	1.17	28.0	+0.10	1.10	35.4	+0.20	196 / 196	1.14	31.6	+0.15	43 / 17
SD3.5-Large + IP-Adapter†	0.93	40.0	-0.20	1.31	22.0	-0.16	74/74†	1.23	25.7	-0.17	10 / 19

† SD3.5-Large has much lower scored coverage than the others, especially in S4, so its results should be interpreted cautiously.

Table 7: **Task 1 correction on common-object generations.** Metrics are computed on matched scorable pairs before and after *Rescale* correction. Short names are used for compactness.

Metric	Gemini	GPT	Z-Image	Grok	Qwen	FLUX	SD3.5
Error before ↓	**0.592**	0.692	0.759	0.785	0.916	1.025	1.041
Error after ↓	**0.383**	0.445	0.431	0.450	0.423	0.529	0.635
Gain by *Rescale* ↑	+0.208	+0.246	+0.328	+0.335	+0.493	**+0.495**	+0.406

Table 8: **Correction results on image-conditioned generation/editing tasks.** Task 2 reports *Rescale* correction on human-product generations from each source model; Task 3 compares *Rescale* with general-purpose editors on the correction benchmark.

Task	Metric	Gemini	GPT	Seedream	Qwen	FLUX	SD3.5	*Rescale*
Task 2	Error before ↓	0.261	**0.231**	0.295	0.306	0.406	0.565	–
	Error after ↓	**0.086**	0.090	0.168	0.144	0.228	0.256	–
	Gain by *Rescale* ↑	+0.175	+0.141	+0.126	+0.162	+0.178	**+0.309**	–
Task 3	Error before ↓	1.263	1.275	1.280	1.259	1.247	**1.042**†	1.258
	Error after ↓	0.974	0.674	1.130	1.127	1.121	1.208	**0.548**
	Gain ↑	+0.289	+0.601	+0.150	+0.132	+0.126	-0.167	**+0.710**

†SD3.5 has substantially lower matched coverage on Task 3, so its before-edit error is not directly comparable to other editors.

model, suggesting that common-object mean-regression and human-product metric-scale errors are often locally correctable. On Task 3, *Rescale* yields the largest gain over the same correction inputs, showing that explicit scale diagnosis and structured local editing are more effective than generic image editing. Overall, *GenScale*'s physical-size structure is not only evaluative but actionable, though autonomous scale repair remains imperfect.

Identity and Visual-quality Preservation. The ideal scale correction should not trade geometric plausibility for image degradation, we evaluate paired pre- and post-correction images in Tab. 9. We use CLIP-I [49] and DINO [44] for visual/identity consistency, SSIM [63] and SSIM-HF [63, 32] for image-level and high-frequency preservation, and LAION-Aes [56] and Q-Align-IQ [67] for no-reference visual quality. CLIP-I, DINO, SSIM, and SSIM-HF remain high despite the intended object-size change, which naturally lower pairwise similarity because scale itself is part of the visual semantics. Meanwhile, LAION-Aes and Q-Align-IQ change by at most 1.7% and 1.9%, respectively, indicating that *Rescale* improves relative-scale plausibility with negligible degradation in image quality and aesthetics. Bootstrap confidence intervals are reported in Sec. F

6 Conclusion

In summary, we introduced *GenScale*, a benchmark for relative object scale in image generation and editing, covering common-object scale priors, human-product metric scale, and scale correction from failed generations. We developed a human-calibrated ordinal evaluator that grounds pairwise scale judgments in object-size metadata, enabling scalable assessment beyond raw pixel ratios. We further introduced *Rescale*, a model-agnostic post-generation correction agent that uses structured scale information for localized geometry-aware editing. Experiments on contemporary open and closed models show that relative scale remains unreliable, while *Rescale* consistently improves scale plausibility. Together, these results establish relative object scale as a distinct, measurable, and actionable dimension of physical realism.

Limitations and Future Work. *GenScale* targets relative object scale rather than general physical or spatial realism, and its knowledge base covers visually identifiable categories with stable physical dimensions. It therefore excludes deformable, fine-grained, or context-dependent objects, and needs broader validation across viewpoints, occlusions, domains, and future model families. *Rescale* assumes scale-relevant objects are visible and locally editable; future work could incorporate metric size priors or scale-aware preference learning directly into generative models.

Table 9: **Identity preservation and visual quality after correction.** Higher values indicate better preservation or quality; percentages denote relative change after correction.

Setting	CLIP-I ↑	DINO ↑	SSIM ↑	SSIM-HF ↑	LAION-Aes (before / after) ↑	Q-Align-IQ (before / after) ↑
Task 1	94.8	90.2	88.8	92.4	5.83 / 5.73 (−1.7%)	4.74 / 4.67 (−1.5%)
Task 2	92.4	84.5	73.5	82.1	4.99 / 4.96 (−0.8%)	4.88 / 4.88 (0.0%)
Task 3	95.6	88.9	89.3	92.9	5.83 / 5.73 (−1.2%)	4.76 / 4.67 (−1.9%)

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Supplementary Material

Overview

This appendix is organized as follows:

Sec. A describes the LLM/VLM components used in *GenScale*, including the Gemini-based scale judge and the multimodal planning agent in *Rescale*.

Sec. B reports the compute resources, API usage, and runtime required for benchmark generation, automatic evaluation, human calibration, and *Rescale* correction experiments.

Sec. C provides additional details of *GenScale* construction, including physical-size metadata, object and product filtering, prompt construction, task-specific sampling rules, Task 3 failure-case selection, and the released metadata format. This section supports Sec. 3.1 and Sec. 3.2 of the main paper.

Sec. D gives full details of the human-calibrated scale evaluation protocol, including the human annotation interface, annotator instructions, Gemini judge prompts, repeated-query aggregation, validity criteria, calibration diagnostics, and bootstrap confidence intervals. This section supports Sec. 3.3 of the main paper.

Sec. E describes *Rescale* in more detail, including agentic scale diagnosis, edit-plan construction, object localization, segmentation, background completion, depth conditioning, local insertion, multi-round verification, insertion-backend training, and ablation studies. This section supports Sec. 4 of the main paper.

Sec. F reports the complete quantitative results for GenScale Tasks 1–3 and *Rescale* correction, including full score distributions, directional error breakdowns, valid-pair coverage, matched before/after metrics, statistical confidence intervals, and quality-preservation metrics. This section supports Sec. 5.1 and Sec. 5.2 of the main paper.

Sec. G presents additional qualitative examples across all GenScale tasks, including generations from evaluated models, correction results from general-purpose editors, *Rescale* before/after examples, and representative failure cases.

Sec. H provides a direct comparison between *GenScale* and GenSpace, clarifying how *GenScale* differs in its focus on real-world relative object scale, human-product metric scale, and post-generation scale correction.

Sec. I discusses benchmark release format, asset licenses, responsible release safeguards, human-subject details, broader impacts, limitations, and ethical considerations.

A LLM/VLM Usage Details

This work uses LLM/VLM-based components as part of the research methodology, beyond writing or formatting assistance. We describe their roles here for transparency.

First, GenScale uses a Gemini-based VLM judge for scalable pairwise relative-scale evaluation. For each anchor–target pair, the judge is given the generated image, the anchor and target object names, their reference physical lengths, the expected 3D length ratio, and the five-point ordinal scale rubric described in Sec. 3. When available, the original generation prompt is provided only to disambiguate the intended objects or layout, not as evidence that the rendered scale is correct. The VLM judge is not used as an unvalidated source of ground truth: it is calibrated against human annotations from nine raters, and we report exact agreement, within-one agreement, MAE, Pearson correlation, and quadratic-weighted kappa against human consensus. For automatic scoring, each scorable pair is queried five times and the final score is obtained by majority vote, with median tie-breaking when no unique mode exists. Images failing the predefined visibility and validity criteria are excluded from scored-pair metrics, while ordinary scale errors remain scored.

Second, *Rescale* uses a multimodal agent to convert image evidence and structured physical-size metadata into localized scale-correction plans. Given an image, object identities, physical-size references, and detected or provided object boxes, the agent selects an editable target object and a fixed anchor object, estimates a conservative resize factor, and predicts a contact-preserving anchor point for local reinsertion. The agent's output is an edit plan specifying the target, anchor, bounding box, resize factor, and anchor point; the actual image modification is then executed by a modular editing pipeline using segmentation, background completion, depth estimation, masking, and insertion.

Table 10: **Compute and API usage summary.** The table reports the main compute components used in the final experiments. Gemini API components use Gemini-3.1-Pro-preview [17] and Gemini-3.1-Flash-Image-preview [18]. API cost is approximate because calls were made under shared credits and exact billing was not logged.

Component	Main model / hardware	Scale of final run	Notes
Benchmark prompt construction	Gemini-3.1-Pro-preview API	900 benchmark entries: 400 Task 1, 300 Task 2, 200 Task 3	Used to synthesize evaluation prompts and Task 3 correction instructions; physical-size metadata was not produced by the VLM.
Benchmark image generation / editing	Public generation and editing APIs	Task 1: 7 models $\times$ 400 prompts; Task 2: 6 models $\times$ 300 prompts; Task 3: 6 models $\times$ 200 edit prompts	Approximately 5,800 final generated or edited images, before minor reruns for failed API calls.
Automatic VLM judge	Gemini-3.1-Pro-preview API	Five repeated judge calls per scorable pair	Majority vote is used; if no unique mode exists, the median score is used.
Rescale inference	Modal GPU workers; A100 used in our main runs	Batch inference on Task 1 and Task 2 correction sets	The pipeline also runs on NVIDIA L40S and RTX A6000; at least 36GB VRAM is sufficient with CPU/offload options.
Insertion-backend training	Single NVIDIA H200 GPU	500K optimization steps; approximately 10 days	Training uses InsertAnything initialization, FLUX Fill / Redux, depth ControlNet conditioning, and LoRA-based fine-tuning.

The agent is also used for post-edit verification in multi-round correction, where an additional round is applied only when an obvious scale inconsistency remains.

LLM/VLM outputs are not used to define the benchmark's physical-size metadata or product dimensions. Those quantities are derived from external object-size sources, product metadata, and human-anchor specifications as described in Sec. 3. Similarly, the VLM judge is fixed before full benchmark evaluation and is applied uniformly to all evaluated generators, editors, and *Rescale* outputs. The full judge prompts, edit-planning prompts, aggregation rules, and validity criteria are provided in the appendix to support reproducibility.

B Compute Resources and Reproducibility Details

This section summarizes the compute resources used for benchmark construction, automatic evaluation, *Rescale* inference, and insertion-backend training. All large-scale image generation and VLM-based scoring experiments were run through public model APIs or Modal GPU workers. Unless otherwise noted, benchmark construction, automatic judging, and agentic planning used Gemini-3.1-Pro-preview [17]. For object removal inside *Rescale*, we used Gemini-3.1-Flash-Image-preview [18]. Exact monetary API cost was not centrally logged because the experiments were run under shared institutional credits; based on the number of generated images, edited images, and VLM judge calls, the API cost for the reported benchmark generation and evaluation was below approximately $500, excluding unused exploratory runs.exploratory runs.

B.1 Benchmark Generation Compute

Benchmark construction consists of prompt construction, metadata assembly, and Task 3 failure-case selection. Task 1 and Task 2 prompts are generated with Gemini-3.1-Pro-preview [17] using structured physical-size metadata as input. For Task 1, the script samples two to four non-human common objects from the physical-size knowledge base and keeps only object sets whose largest-to-smallest characteristic-length ratio lies in $[4, 20]$. For Task 2, the script selects one ABO product [9] and

Table 11: **Benchmark construction workload.** Prompt construction uses Gemini-3.1-Pro-preview, while image generation and editing use the evaluated image-generation or editing models.

Task	Entries	Pairwise relations	Evaluated models	Main workload
Task 1	400	1,143	7	Text-to-image generation from common-object prompts
Task 2	300	300	6	Product-reference generation with explicit product dimensions and human anchors
Task 3	200	200	6	Image editing from erroneous Task 1/2 generations
Total	900	1,643	–	Approximately 5,800 final generated or edited model outputs

assigns a human anchor according to product characteristic length. The final benchmark contains 400 Task 1 entries, 300 Task 2 entries, and 200 Task 3 entries.

Task 3 is built from previously generated Task 1 and Task 2 images with non-perfect scale scores. The construction script first filters candidate images by scale score, then optionally applies VLM-based visual quality control to remove cases with missing objects, duplicate primary objects, severe blur, or synthesis failures that make correction ill-defined. Each retained source image is converted into two edit settings: S4, where the model must diagnose and correct the scale error, and S5, where the target object and scale factor are provided.

B.2 Automatic Judge Compute and API Usage

Automatic evaluation uses Gemini-3.1-Pro-preview [17] as the calibrated VLM judge. For each scorable anchor–target pair, we issue five independent judge calls and aggregate the five ordinal scores by majority vote, using the median score when no unique mode exists. The same scene-level validity prefilter is applied before pairwise scoring; invalid images are excluded from scored-pair metrics but are still reflected in valid-image coverage.

The final benchmark evaluation involves approximately 55K pairwise judge calls before retry overhead:

$$7 \times 1{,}143 \times 5 + 6 \times 300 \times 5 + 6 \times 200 \times 5 = 55{,}005.$$

This estimate counts only the final reported model outputs and repeated pairwise judge calls. It excludes retries from transient API errors, prefilter calls, prompt-construction calls, exploratory experiments, and human-calibration runs. The evaluation scripts support request pacing, per-request timeouts, retry logic with exponential backoff, image-level parallelism, pair-level parallelism, and repeated sampling per pair. For the final reported numbers, we used the same judge model and aggregation rule across all evaluated generators, editors, and *Rescale* outputs.

B.3 Human Calibration Cost and Annotation Time

Human calibration was used only to validate the automatic judge, not to score the full benchmark. Nine annotators labeled a calibration subset of 285 images and 527 pairwise object relations. Annotators were shown the generated image, anchor and target object names, physical reference lengths, the expected 3D length ratio, and the five-point ordinal scale rubric. They also marked invalid cases where the relevant objects were missing, duplicated, merged, or too ambiguous for reliable pairwise scale judgment.

We did not use human annotations to tune model-specific scores after automatic evaluation. The Gemini judge prompt and aggregation protocol were fixed before full benchmark scoring. Additional details on the annotation interface, consensus construction, and inter-rater agreement are provided in Sec. D.

B.4 *Rescale* Correction Runtime

Rescale inference was run on Modal GPU workers. Our main inference runs used A100 GPUs, but the pipeline also runs on NVIDIA L40S and RTX A6000 GPUs. A GPU with at least 36GB VRAM is sufficient when model caching and CPU offload options are enabled. The Modal inference scripts allocate one GPU worker per shard and reuse loaded models within a worker to avoid repeated model initialization.

Each correction case runs a modular pipeline consisting of agentic diagnosis, target localization, object extraction, segmentation, background completion or object removal, depth processing, local insertion, blending, and optional verification. The agentic steps use Gemini-3.1-Pro-preview [17]

Table 12: ***Rescale* inference configuration.** The reported values are the main default settings used by the Modal inference scripts for the final correction experiments.

Component	Configuration
Main GPU used in our runs	NVIDIA A100
Other tested GPUs	NVIDIA L40S and RTX A6000
Minimum practical VRAM	Approximately 36GB with offload/caching options
Agent / planning VLM	Gemini-3.1-Pro-preview
Object remover	Gemini-3.1-Flash-Image-preview
Insertion denoising steps	30
Guidance scale	30.0
Depth ControlNet scale	0.6
Depth ControlNet end fraction	0.6
Batch execution	Modal sharding by image range; loaded models reused within each worker
Resume behavior	Completed cases are skipped if the final output marker already exists

for scene-level planning, object grounding, bounding-box refinement, and verification. Gemini-3.1-Flash-Image-preview [18] is used as the image-based object remover inside the correction pipeline. Segmentation and depth processing use SAM2 [52] and Depth Anything V2 [70], respectively. The image insertion stage uses the trained insertion backend described below, with 30 denoising steps, guidance scale 30.0, depth ControlNet [74] scale 0.6, and ControlNet ending fraction 0.6 in the main inference configuration.

For batch inference, the scripts support sharding by image index range, resume-by-output-marker, and multi-container execution. Task 1 inference can run in a multi-round mode that edits multiple detected scale errors sequentially within the same worker, reusing loaded models between rounds. Task 2 inference uses a product-human-specific controller with scene prefiltering, residual scale correction, depth-aware mask dilation, and conservative scale clamps.

B.5 Training / Fine-tuning Compute for the Insertion Backend

The insertion backend is trained as a modular component of *Rescale* and is not the core benchmark contribution. It can be replaced by other insertion models, as shown in the ablations in Sec. E. Our trained backend starts from the InsertAnything LoRA checkpoint [57] and adapts a FLUX-based insertion model for scale correction. The architecture uses FLUX.1-Fill-dev as the inpainting backbone, FLUX.1-Redux-dev [3] as the reference-image encoder, and FLUX.1-dev-ControlNet-Depth as the depth-conditioning branch. The trainable component is a LoRA adapter [24] with rank 256; the training also enables depth conditioning through ControlNet-style conditioning [74] and high-frequency injection.

Training was run on a single NVIDIA H200 GPU for 500K optimization steps and took approximately 10 days. The launch script sets batch size 1 and gradient accumulation 2, giving an effective batch size of 2 on the single-GPU run. We use gradient checkpointing, bfloat16 model weights, four dataloader workers, checkpoint saving every 1K steps, sample visualization every 200 steps, and logging every 20 steps. Although the training launch script supports longer runs, the checkpoint used in the reported experiments was trained for 500K steps.

C *GenScale* Benchmark Details

C.1 Physical-Size Knowledge Base

GenScale is grounded in a physical-size knowledge base that stores characteristic object lengths and, when available, plausible length ranges. For common-object scale reasoning, we start from visually identifiable categories in COCO [36] and LVIS [20], and remove categories whose physical extent is highly variable, visually ambiguous, or non-rigid. For product-scale reasoning, we use product-level dimensions and reference images from Amazon Berkeley Objects (ABO) [9]. Human-anchor dimensions are grounded in ANSUR II anthropometric measurements [19]. Additional object dimensions are derived from domain-specific references, including Animal Diversity Web [41], USDA FoodData Central [61], ISO and ANSI standards [30, 29, 1], traffic specifications [12], and official sport-object specifications [38]. The goal is not maximal category coverage, but stable and externally grounded physical extent for reliable relative-scale evaluation.

Table 13: **Insertion-backend training configuration.** The insertion backend is a replaceable module in *Rescale*; the table reports the configuration used for our trained backend. Training data are drawn from SAM/SA-1B [33], LVIS [20], saliency datasets [5, 62], YouTube-VOS [69], VIPSeg [40], MOSE [10], VITON-HD [8], and AnyInsertion/InsertAnything data [57].

Item	Configuration
Hardware	1 NVIDIA H200 GPU
Training duration	Approximately 10 days
Optimization steps	500K steps for the reported checkpoint
Batch size / gradient accumulation	1 / 2
Effective batch size	2
Precision	bfloat16
Base inpainting model	FLUX.1-Fill-dev
Reference encoder	FLUX.1-Redux-dev
Depth conditioning model	FLUX.1-dev-ControlNet-Depth
Initialization	InsertAnything pretrained LoRA
Trainable adapter	LoRA, rank 256
Depth ControlNet scale	0.6
Depth dropout probability	0.5
Gradient checkpointing	Enabled
Dataloader workers	4
Checkpoint interval	1K steps
Visualization interval	200 steps
Logging interval	20 steps
Training datasets	SAM, LVIS, Saliency, YoutubeVOS, VIPSeg, MOSE, VitonHD, AnyInsertion

Tab. 14 summarizes the semantic and source coverage of the resulting knowledge base. The 100 entries include 97 non-human common objects used for Task 1 and common-object correction cases in Task 3, together with three human body-part anchors used for Task 2.

C.2 Task 1: Common-Object Sampling and Prompt Templates

Task 1 evaluates implicit common-object scale priors in text-to-image generation. Each entry contains only a text prompt with two to four non-human objects; no object-specific metric size or reference image is provided to the generator. Objects are sampled from the 97 non-human common-object entries in the physical-size knowledge base. We retain sampled sets whose largest-to-smallest characteristic-length ratio lies in $[4, 20]$, which removes near-equal comparisons that are difficult to judge and extremely disparate combinations that often become layout-dominated rather than scale-diagnostic.

Tab. 16 summarizes the sampling protocol, and Tab. 15 reports the final object-count distribution. For an n-object prompt, we evaluate all $\binom{n}{2}$ unordered object pairs, so multi-object prompts increase pair-level evaluation density without increasing the number of generated images. Task 1 contains 400 prompts and 1,143 evaluated relations, split nearly evenly between S1 Natural Depth and S2 Same Plane.

To extend the analysis from Fig. 6, we estimate object depth for every Task 1 generated image using Depth Anything V2. Because bounding boxes are not segmentation masks, we avoid full-box averaging: for each object, we use the central 60% of its bounding box, discard the top and bottom 5% depth values, and take the median of the remaining values as the object depth. We then convert object depths into within-image depth ranks, where 0 denotes the nearest object and 1 denotes the farthest. Objects are grouped by their real-world size rank within each image, and we compute both the average depth rank of each size group and the Spearman correlation between the benchmark physical length $\log \ell$ and the estimated depth rank.

The full diagnostic in Tab. 18 confirms that depth placement mainly affects S1. In S1, all seven generators place smaller objects closer and larger objects farther away: on average, the smallest objects have depth rank 0.176, while the largest objects have depth rank 0.863. Equivalently, 79.2% of unequal-size object pairs place the smaller object closer to the camera, and the mean size–depth Spearman correlation is 0.558. In contrast, S2 largely removes this pattern: the average depth ranks of small, middle, and large objects become much closer (0.423/0.480/0.592), the smaller-closer rate drops to 52.2%, and the correlation falls to 0.129. This supports the interpretation that natural depth in S1 can partially hide relative-scale errors through depth-mediated scale compression, whereas S2 exposes scale errors more directly by constraining objects to a similar depth plane.

Table 14: **Coverage of the 100-entry physical scale knowledge base.** Objects are grouped by semantic domain and dimensional source type. The knowledge base contains 42 COCO-derived and 58 LVIS-derived categories, covering both everyday objects and physically standardized objects.

Group	#Obj.	%	Example categories	Dimensional grounding
Human body-part anchors	3	3.0	hand, face, foot	ANSUR II anthropometric measurements [19].
Animals	13	13.0	elephant, giraffe, horse, cat, turtle	Animal Diversity Web and species-level references [41].
Sports / recreation	14	14.0	baseball, basketball, tennis racket, surfboard	Official sport rules and standardized equipment dimensions.
Electronics / appliances / media	10	10.0	keyboard, laptop, TV, microwave, CD	ANSI, ISO, and typical product specifications [29, 1].
Office / personal small objects	16	16.0	battery, credit card, passport, pencil, toothbrush	ISO, ANSI, and common product standards [30, 29, 1].
Tableware / containers	11	11.0	bottle, wine glass, mug, fork, bowl, can	Standard tableware and container dimensions.
Food items	7	7.0	apple, orange, banana, carrot, egg	USDA FoodData Central and common food-size standards [61].
Vehicles / traffic / public infrastructure	12	12.0	bicycle, car, bus, stop sign, fire hydrant	Vehicle, traffic, and infrastructure specifications [12].
Household / tools / accessories / instruments	14	14.0	umbrella, suitcase, hammer, watch, guitar	Common product and instrument dimensions.
Total	100	100.0	–	–

Table 15: **Task 1 object-count distribution.** Each n-object prompt induces all $\binom{n}{2}$ unordered pairwise scale relations. Thus, three- and four-object prompts substantially increase the number of evaluated relations without increasing the number of generated images.

Scenario	2 objects	3 objects	4 objects	Total
S1 Natural Depth	93 images 93 pairs	55 images 165 pairs	52 images 312 pairs	200 images 570 pairs
S2 Same Plane	81 images 81 pairs	74 images 222 pairs	45 images 270 pairs	200 images 573 pairs
Task 1 total	174 images 174 pairs	129 images 387 pairs	97 images 582 pairs	400 images 1,143 pairs

C.3 Task 2: ABO Product Filtering and Human-Anchor Assignment

Task 2 evaluates explicit metric scale realization in human-product image generation. Each entry contains one ABO product [9], one product reference image retrievable from ABO metadata, a text prompt specifying the product dimensions, and one size-conditioned human anchor. We use ABO because it provides product-level dimensions and product images, enabling evaluation of whether a model can translate metric product size into plausible human-product proportions.

We filter ABO products using four criteria. First, we retain products with an English listing title. Second, we require valid item dimensions. Third, we require a valid main image identifier that can be mapped to an available ABO image. Fourth, after converting all available dimensions to centimeters, we use the maximum of length, width, and height as the characteristic product length and retain products whose characteristic length lies in $[5, 250]$ cm. This range removes tiny objects that are difficult to resolve visually and very large products that are unlikely to form a well-controlled

Table 16: **Task 1 sampling protocol.** Task 1 samples 2–4 common objects and filters combinations by physical disparity.

Choice	Implementation
Object pool	97 non-human common objects from the physical-size knowledge base; human body-part anchors are excluded.
Prompt size	Each prompt contains 2–4 objects. The final split contains 174 two-object, 129 three-object, and 97 four-object prompts.
Disparity filter	With $D = \max_i l_i / \min_i l_i$, we keep only combinations satisfying $4 \leq D \leq 20$.
Rationale	Smaller disparities are often hard to judge reliably, whereas larger disparities tend to produce layout-dominated scenes.
Scenario split	S1 allows natural depth and perspective; S2 places objects on a similar depth plane to reduce perspective shortcuts.
Metric cues	No numeric size is shown to the model; the task tests implicit common-object scale priors.

Table 17: **Task 1 prompt templates.** S1 permits realistic perspective and depth variation, while S2 constrains the objects to a similar depth plane.

Scenario	Prompt template
S1: Natural Depth	Create a photorealistic scene containing `[object list]`. The objects should appear naturally in the scene with physically plausible real-world relative sizes.
S2: Same Plane	Create a photorealistic scene containing `[object list]`. Place all objects on approximately the same depth plane, such as on the same tabletop or floor surface, with physically plausible real-world relative sizes.

human-anchor interaction. For each retained product, we store length, width, height, characteristic length, and a narrow product-specific tolerance interval.

Human anchors are assigned deterministically from the product characteristic length. Products shorter than 30 cm are paired with a human hand; products between 30 and 60 cm are paired with a human face/head; products between 60 and 100 cm are paired with a human foot/leg; products longer than 100 cm are paired with a full human body. The corresponding canonical anchor lengths are derived from anthropometric references [19]. Tab. 19 gives the anchor-assignment policy, and Tab. 20 reports the final anchor distribution in the 300-entry Task 2 split.

The Task 2 prompt contains both semantic and metric constraints. The product title is cleaned into a concise noun phrase, while the metric block provides the characteristic length and all available axis-aligned dimensions. The prompt asks the model to depict a natural, purpose-consistent interaction between the product and the assigned human anchor, with catalog-style visibility and mild perspective so that product-to-anchor scale remains interpretable.

C.4 Task 3: Failure-Case Selection and Correction Prompt Construction

Task 3 evaluates scale-error correction rather than average-case image editing. We construct Task 3 from failed but editable outputs in Tasks 1 and 2. A source case is retained only when the main objects are recognizable, the scene is visually usable, and at least one pairwise scale relation is not perfectly plausible under the GenScale ordinal rubric. For Task 1-derived cases, we restrict source images to two-object prompts so that the correction target is unambiguous. For both Task 1 and Task 2 sources, we remove images with missing or unidentifiable objects, duplicate primary objects, severe synthesis artifacts, heavy occlusion, blur, or extreme correction factors that make localized editing ill-defined. This filtering isolates correction of object scale rather than recovery from invalid image generation.

The final Task 3 set contains 100 source images, with 61 from Task 1 and 39 from Task 2. Each source image is converted into two edit settings. In S4 Auto-Discovery, the model receives the erroneous image and a general instruction to check whether object size proportions are unrealistic; if an error exists, it must infer which object should be resized and perform the correction while preserving identity, composition, background, and lighting. For Task 2-derived S4 cases, the prompt additionally includes compact product and human-anchor reference lengths so that the editor can judge the product-human scale relation. In S5 Precise Scale Instruction, the prompt directly specifies

Table 18: **Full Task 1 depth-size diagnostic.** Objects are grouped by real-world size within each generated image. Depth rank is averaged within each size group, where 0 denotes the nearest object and 1 denotes the farthest.

Model	Split	Img.	Obj.	Pairs	Mean depth rank by size bucket			Smaller closer (%)	Size–depth ρ
					Small	Middle	Large		
Gemini Image Preview	S1	199	555	562	0.136	0.452	0.903	80.6	0.620
	S2	196	551	556	0.545	0.493	0.460	41.5	-0.085
	All	395	1106	1118	0.339	0.473	0.683	61.2	0.269
GPT-Image-2	S1	199	555	562	0.148	0.457	0.888	81.5	0.595
	S2	200	564	572	0.501	0.503	0.497	44.1	-0.006
	All	399	1119	1134	0.325	0.480	0.692	62.6	0.294
Qwen-Image 2512	S1	196	542	540	0.164	0.450	0.876	78.1	0.582
	S2	198	557	562	0.342	0.504	0.656	58.0	0.246
	All	394	1099	1102	0.253	0.478	0.765	67.9	0.412
FLUX.2	S1	198	537	519	0.145	0.476	0.872	82.3	0.591
	S2	199	551	543	0.352	0.462	0.678	58.7	0.258
	All	397	1088	1062	0.249	0.469	0.775	70.2	0.422
SD3.5-Large	S1	179	495	495	0.233	0.419	0.830	74.1	0.502
	S2	166	455	442	0.326	0.477	0.692	61.3	0.290
	All	345	950	937	0.278	0.447	0.763	68.1	0.398
Grok Image	S1	198	553	562	0.183	0.452	0.855	79.7	0.536
	S2	200	564	572	0.494	0.473	0.528	47.6	0.021
	All	398	1117	1134	0.340	0.463	0.691	63.5	0.274
Z-Image-Turbo	S1	193	531	526	0.224	0.449	0.816	77.8	0.483
	S2	181	491	468	0.404	0.445	0.635	54.1	0.178
	All	374	1022	994	0.311	0.447	0.728	66.6	0.333
Average	S1	1362	3768	3766	0.176	0.451	0.863	79.2	0.558
	S2	1340	3733	3715	0.423	0.480	0.592	52.2	0.129
	All	2702	7501	7481	0.299	0.465	0.728	65.7	0.343

Table 19: **Human anchor selection in Task 2.** Products are paired with a human anchor according to their characteristic length, enabling interpretable product-to-human scale evaluation across small, medium, large, and oversized products.

Product characteristic length	Human anchor	Anchor length	#Entries
$<$ 30 cm	human hand	19.3 cm	151
30–60 cm	human head/face	24.0 cm	77
60–100 cm	human foot/leg	26.5 cm	31
$\geq$ 100 cm	full human body	170.0 cm	41

the fixed reference object, editable target object, resize direction, numeric scale factor, and target size ratio. This separates fine-grained resize-instruction following from autonomous error diagnosis.

Tab. 22 summarizes the construction protocol, and Tab. 23 summarizes the S4/S5 edit-instruction format.

Tab. 24 summarizes the visual quality check (**QC**) applied when constructing Task 3 from failed two-object examples in Task 1 and Task 2. Most candidates passed QC and were retained, yielding 61 Task 1 sources and 39 Task 2 sources. The rejected cases were mainly due to missing or visually unclear target objects, while duplicate-object ambiguity and severe synthesis failures were less frequent.

C.5 Pairwise Ratio Definition and Metadata Schema

Each GenScale entry is an image-level prompt or editing instance, but evaluation is performed at the pair level. For an anchor-target pair (a, t), let l_a and l_t denote their characteristic physical lengths. We define the expected target-to-anchor physical ratio as

$$r_{t/a} = \frac{l_t}{l_a}.$$

When lower and upper physical ranges are available, we also store a tolerance interval

$$\left[\frac{l_t^{\min}}{l_a^{\max}}, \frac{l_t^{\max}}{l_a^{\min}}\right],$$

Table 20: **Human-anchor distribution in Task 2.** Each Task 2 prompt contains one product and one human anchor, resulting in one evaluated product-to-anchor scale relation per image. Product length is computed from `product_scale.typical_len_cm`. Ratio denotes the mean ground-truth target-to-anchor length ratio.

Human anchor	Entries	%	Median length	Length range	Ratio
Human hand	151	50.3	18.29 cm	5.08–30.23 cm	0.94
Human head / face	77	25.7	39.62 cm	30.00–58.42 cm	1.72
Human foot / leg	31	10.3	70.61 cm	31.75–96.52 cm	2.81
Full human body	41	13.7	150.00 cm	100.00–243.84 cm	0.95
Total	300	100.0	–	–	–

Table 21: **Task 2 prompt-construction constraints.** Task 2 combines an ABO product reference, explicit metric product dimensions, and a size-conditioned human anchor.

Component	Construction rule
Product scale block	Provide the characteristic product length and all available length/width/height dimensions in centimeters. The prompt emphasizes that the generated product must match this order of magnitude rather than infer size from the title alone.
Human anchor	Select hand, face/head, foot/leg, or full body according to the product characteristic length. The anchor is fixed for metric consistency.
Interaction constraint	Ask for a natural, purpose-consistent interaction between the product and the human anchor, avoiding arbitrary or decorative placements that ignore product use.
Visibility constraint	Require catalog-style framing where the product remains recognizable and largely unobstructed; mild contact or partial occlusion by the human anchor is allowed.
Camera constraint	Favor front or three-quarter mild-perspective views rather than extreme wide-angle compositions, so apparent product-to-anchor ratios remain interpretable.

which accounts for intra-class variation and measurement uncertainty. For Task 2, product dimensions are taken from ABO metadata [9], while human-anchor lengths are fixed canonical measurements derived from anthropometric references [19]. For Task 3, the pairwise ratio is inherited from the corresponding Task 1 or Task 2 source case.

Tab. 25 summarizes how prompts, scale information, and ground-truth ratios differ across the three tasks.

C.6 Benchmark Release Format

We release *GenScale* as an image-level benchmark specification with associated pairwise scale records. The main benchmark file, `GenScale_Benchmark.json`, contains 900 entries spanning Task 1–3. Each entry specifies a stable task identifier, scenario label, model input condition, object list, physical-size metadata, and one or more ground-truth pairwise scale ratios. For reproducibility, we additionally release the physical-size knowledge base, the sampled ABO product metadata used for Task 2 construction, the erroneous source images used by Task 3, and the evaluation scripts used to score model outputs. Task 2 product reference images are not redistributed as a separate image package; they can be obtained from the ABO dataset using the released ABO metadata and image identifiers.

The JSON schema is task-dependent but follows a common structure. Task 1 entries contain a text prompt, a scenario label, the included common objects, and all unordered pairwise ratios induced by the object set. Task 2 entries add ABO-derived product metadata and an ABO image identifier for retrieving the product reference image. We do not redistribute ABO product images directly; instead, the released metadata specifies the selected product/image records so that users can obtain the same references from the ABO dataset under its original terms. Task 3 entries contain an erroneous source image, the correction prompt, source-task provenance, and the pairwise ratio inherited from the corresponding Task 1 or Task 2 source case. Unlike Task 2 product references, Task 3 source images are released with the benchmark because they are generated failure cases and serve as the direct input to the editing task. For S5, the entry additionally stores an explicit edit plan containing the target object, reference object, resize direction, and scale factor.

Table 22: **Task 3 construction protocol.** Task 3 is built from failed generations in Task 1 and Task 2, and each selected source image is converted into two correction prompts.

Component	Description
Source images	Failed or imperfect generations from Task 1 and Task 2. In the final benchmark, Task 3 uses 100 source images: 61 from T1 and 39 from T2.
Source models	T1 source images are generated by FLUX.2; T2 source images are generated by Qwen-Image.
Quality control	We remove images with severe synthesis failures, missing/unidentifiable target objects, duplicate main objects, or heavy occlusion/blur that would make scale correction ill-defined. Selection is not based on downstream editing success.
S4: Hard auto-discovery	The model is given the erroneous image and reference scale information, and must decide whether a correction is needed and which object to resize.
S5: Precise scale instruction	The model is directly given the object to resize and the exact scale factor, isolating scale-instruction following from error diagnosis.
Expansion	Each source image yields two prompts, one S4 and one S5, producing 200 Task 3 entries.

Table 23: **Task 3 correction-prompt construction.** Each retained erroneous source image is converted into both an auto-discovery correction prompt and a precise scale-instruction prompt.

Scenario	Construction rule
S4: Auto-Discovery	Ask the editor to inspect the image for unrealistic object-size proportions. If a scale error exists, the editor must automatically identify the necessary object-size correction and preserve object identity, pose, viewpoint, composition, background, and lighting.
S4 from Task 2	Use the same auto-discovery instruction, but also provide compact product and human-anchor reference lengths, because Task 2 errors depend on explicit product-human metric scale.
S5: Precise Scale Instruction	Provide a strict edit instruction specifying the object to keep unchanged, the object to edit, the resize direction, the exact multiplicative scale factor, and the target pairwise ratio. The prompt forbids identity, position, background, lighting, or viewpoint changes.

To evaluate a new model, users generate images using the released prompts and reference images, save outputs under the expected task identifiers, and run the provided task-specific evaluation scripts. The evaluator reads the corresponding pairwise records from `GenScale_Benchmark.json`, applies the scene-level validity prefilter, queries the calibrated VLM judge for each scorable pair, and aggregates pairwise ordinal scores by task, scenario, and model. Task 3 evaluation reuses the Task 1 or Task 2 scoring logic according to each entry's `source_task_type`, ensuring that edited images are judged with the same criterion as their original source case.

D Human and VLM Evaluation Details

D.1 Human Annotation Interface and Instructions

We built a custom annotation GUI for pairwise relative-scale evaluation, shown in Fig. 7. Each screen contains a generated image on the left and pairwise evaluation cards on the right. A single image may induce multiple anchor–target pairs, especially in Task 1 multi-object prompts; annotators score each pair independently while looking at the same image.

For each pair, the interface displays the *anchor* object, the *target* object, and their reference physical lengths in centimeters. Annotators are instructed to treat the anchor as correctly sized and judge only whether the target appears too small, proportionate, or too large relative to that anchor. This role assignment is fixed by the benchmark metadata and must not be reversed. The reference lengths shown in the interface are real-world typical lengths from the physical-size metadata or product catalog; they are intended as semantic scale references rather than pixel measurements in the rendered image.

The interface also records pair-level validity information. Annotators mark whether both objects are clearly visible, whether duplicate instances make the pair ambiguous, and whether merged objects

Table 24: **Task 3 visual quality control statistics.** We report the number of source candidates before and after QC, together with rejection reasons.

Source task	Candidates	Kept	Missing object	Duplicate object	Severe synthesis failure
T1	75	61	8	4	2
T2	42	39	2	0	1

Table 25: **Prompt and ground-truth construction.** *GenScale* separates implicit common-object scale priors from explicit metric scale control and scale-error correction.

Aspect	Description
T1: Common objects	
Input	Text-only generation with two to four common objects.
Scale prompt	No object-specific numeric size is provided; prompts only request physically accurate real-world proportions.
Ground truth	For objects i, j, the expected ratio is $r_{ij} = l_i/l_j$, with acceptable interval $[l_i^{\min}/l_j^{\max}, l_i^{\max}/l_j^{\min}]$.
Evaluation target	Tests whether the model's implicit real-world object-size prior is accurate.
T2: Human–product	
Input	Product reference image plus text prompt.
Scale prompt	Product dimensions from ABO metadata are explicitly included; the human anchor has a fixed canonical size.
Ground truth	The product-to-anchor ratio is computed from the ABO characteristic product length and the assigned human-anchor length, with tolerance intervals.
Evaluation target	Tests whether the model can realize explicit metric product scale while preserving product identity and natural human interaction.
T3: Scale correction	
Input	Erroneous generated image plus edit instruction.
Scale prompt	S4 asks the model to diagnose whether and how to resize; S5 specifies the target object, reference object, resize direction, and scale factor.
Ground truth	The pairwise ratio is inherited from the corresponding T1 or T2 source case.
Evaluation target	Tests whether an editing model can diagnose and correct scale errors without changing identity, pose, or scene composition.

or severe artifacts prevent reliable scale judgment. Navigation controls allow annotators to save the current image-level ratings, move backward, skip unusable examples, or jump to a specific index or task identifier. The calibration interface supports Task 1 scenarios S1 and S2 and Task 2 scenario S3, with configurable scenario ranges and per-scenario quotas.

D.2 Pairwise Ordinal Rubric and Invalid-Pair Criteria

GenScale uses a five-point ordinal rubric rather than a raw pixel-ratio metric because apparent image size depends on object identity, real-world extent, camera perspective, depth ordering, foreshortening, occlusion, and partial visibility. For each valid anchor–target pair, annotators assign a score $s \in \{1, 2, 3, 4, 5\}$:

- $s = 1$: the target is severely undersized relative to the anchor.
- $s = 2$: the target is slightly undersized relative to the anchor.
- $s = 3$: the target has a physically plausible size relative to the anchor.
- $s = 4$: the target is slightly oversized relative to the anchor.
- $s = 5$: the target is severely oversized relative to the anchor.

Operationally, score 3 corresponds to an inferred target-size error within approximately $\pm 20\%$ of the reference scale, scores 2 and 4 correspond to errors between 20% and 60%, and scores 1 and 5 correspond to errors larger than 60%. These thresholds are used as perceptual guidance rather than exact pixel-caliper rules.

Table 26: **Benchmark metadata schema.** Each image-level entry contains task-level information and one or more pairwise scale records used for evaluation.

Field	Description
`entry_id`	Unique image-level benchmark identifier.
`task` / `scenario`	Task label and scenario label, e.g., T1/S1, T1/S2, T2/S3, T3/S4, or T3/S5.
`prompt`	Generation or editing prompt shown to the model.
`objects`	Object names appearing in the prompt or source image.
`product_image`	Product reference image path for Task 2 entries; absent for Task 1.
`source_image`	Erroneous source image path for Task 3 entries.
`physical_lengths`	Characteristic physical lengths and optional lower/upper ranges for each object or anchor.
`pairs`	List of anchor-target records used as the atomic evaluation units.
`pairs.anchor` / `pairs.target`	Reference object and evaluated target object.
`pairs.expected_ratio`	Expected target-to-anchor physical ratio $r_{t/a}$.
`pairs.ratio_interval`	Optional plausible physical-ratio interval induced by object length ranges.
`edit_metadata`	For Task 3, editable object, fixed reference object, resize direction, and scale factor when available.

Table 27: **Released benchmark artifacts.** The benchmark is released as metadata, Task 3 source images, and evaluation code. Paths in the public release are stored relative to the release root rather than as local absolute paths.

Artifact	Format	Content
`GenScale_Benchmark.json`	JSON	Final 900-entry benchmark specification, including prompts, task/scenario labels, object lists, and pairwise target ratios.
`authoritative_kb_3d_100.csv`	CSV	Physical-size knowledge base with 100 object categories and characteristic 3D length statistics used to construct Task 1 and evaluate pairwise ratios.
`abo_local_sampled_1000_representative.csv`	CSV	Intermediate Task 2 product candidate metadata sampled from ABO [9]; used for product filtering, prompt construction, and retrieving the corresponding ABO product reference images.
`source_images/task3/`	Images	Erroneous source images used as editing inputs for Task 3. These images are released because Task 3 cannot be reconstructed from prompts alone.
`eval/`	Python	Task-specific VLM-evaluation scripts for Task 1, Task 2, and Task 3, including scene prefiltering, repeated judge sampling, score aggregation, and output serialization.
`README.md`	Markdown	File-structure description, model-output naming convention, ABO image retrieval instructions for Task 2, and example evaluation commands.

Annotators are explicitly instructed to account for perspective and depth before assigning a score. For example, a target farther from the camera may appear smaller in image space without being physically undersized, and a foreground target may appear larger without being physically oversized. A penalty is assigned only when the target still appears implausibly small or large after considering the likely 3D layout, object contact, foreshortening, and partial visibility. In S2 same-plane cases, the default assumption is that objects lie at approximately similar depth, so visible size differences provide stronger evidence of real-world scale errors unless the image clearly contradicts the prompt.

A pair is treated as invalid only when reliable scale judgment is not possible. Invalid cases include missing or unidentifiable anchor/target objects, very blurry objects, fused or merged objects, severe generation artifacts, or duplicate object instances that make it ambiguous which instance should be scored. Visible but incorrectly scaled objects are not invalidated; they remain scorable and contribute to scale metrics.

Table 28: **Core fields in the released benchmark JSON.** All paths are relative to the release root. Task-specific fields are present only when applicable.

Field	Tasks	Description
`task_id`	T1–T3	Stable entry identifier, e.g., `T1_0000`, `T2_0000`, or `T3_0000`.
`scenario`	T1–T3	Scenario label: S1/S2 for Task 1, S3 for Task 2, and S4/S5 for Task 3.
`prompt`	T1–T3	Text prompt given to the generator or editor.
`objects_included`	T1–T3	Canonical object names expected in the image.
`gt_ratios`	T1–T3	Dictionary of anchor–target pair records. Each record stores the expected length ratio and an acceptable physical-size range.
`reference_image_path`	T2	Relative path to the product reference image used as the image condition.
`product_scale`	T2	Product-level physical dimensions, including typical, minimum, and maximum characteristic lengths in centimeters.
`raw_listing_title`	T2	Original or lightly normalized product title used to identify the product category and dimensions.
`source_task_id`	T3	Identifier of the failed Task 1 or Task 2 entry from which the correction case is derived.
`source_task_type`	T3	Source task type, either `T1` or `T2`; used to dispatch Task 3 outputs to the corresponding evaluator.
`image_path`	T3	Relative path to the erroneous source image used as the editing input.
`prompt_type`	T3	Correction setting: `hard_auto_discovery` for S4 or `precise_scale_instruction` for S5.
`edit_plan`	T3-S5	Explicit correction metadata containing target object, reference object, resize direction, and scale factor.

Table 29: **Per-annotator agreement with the aggregate human consensus.** For each annotator, the reference is the aggregate human consensus constructed from all available raters using modal score aggregation with median-rounded tie breaking. Annotator identities are anonymized.

Annotator	Pairs	Exact (%) ↑	≤ 1 (%) ↑	MAE ↓	QWK ↑	r ↑	Mean diff.
Annotator A	506	70.95	95.26	0.3478	0.8087	0.8181	+0.0040
Annotator B	472	64.19	94.28	0.4280	0.7125	0.7183	-0.0593
Annotator C	519	65.51	94.99	0.4027	0.7853	0.7931	-0.0366
Annotator D	517	70.41	99.03	0.3056	0.8204	0.8404	-0.0387
Annotator E	522	61.88	90.42	0.4828	0.6006	0.6193	-0.0460
Annotator F	513	70.18	96.10	0.3411	0.7947	0.7955	+0.0331
Annotator G	511	55.19	90.02	0.5519	0.7433	0.7700	-0.0039
Annotator H	516	54.26	90.12	0.5562	0.6796	0.6813	+0.0252
Annotator I	526	73.76	96.01	0.3023	0.8371	0.8396	-0.0133
Mean $\pm$ std	511 ± 16	65.15 ± 6.98	94.03 ± 3.16	0.4132 ± 0.0989	0.7536 ± 0.0773	0.7640 ± 0.0760	–
Min–Max	472–526	54.26–73.76	90.02–99.03	0.3023–0.5562	0.6006–0.8371	0.6193–0.8404	–

D.3 Human Consensus Construction

We collect human annotations on a calibration split sampled from Task 1 and Task 2, covering S1 natural-depth common-object scenes, S2 same-plane common-object scenes, and S3 human–product scenes. Task 3 is not separately annotated because each Task 3 case inherits a single pairwise relation from a Task 1 or Task 2 source case, and its edited output can be evaluated using the same source-task criterion.

For each pair, we aggregate all available valid human scores into a consensus label. Let $\mathcal{R}_p$ be the multiset of valid ordinal scores assigned to pair p. The consensus score c_p is the modal score in $\mathcal{R}_p$. When multiple scores are tied for the mode, we break the tie by taking the median of the tied modal scores and rounding to the nearest ordinal label. This preserves the ordinal nature of the rubric and avoids imposing an artificial continuous scale.

We report two forms of human reliability. First, in Tab. 29, each annotator is compared with the aggregate consensus constructed from all available raters. This measures agreement with the final calibration target but includes the evaluated annotator in the reference. Second, in Tab. 30, we rebuild the consensus from the remaining eight annotators and compare the held-out annotator against this leave-one-rater-out reference. This removes self-inclusion bias and gives a more conservative estimate of human reliability.

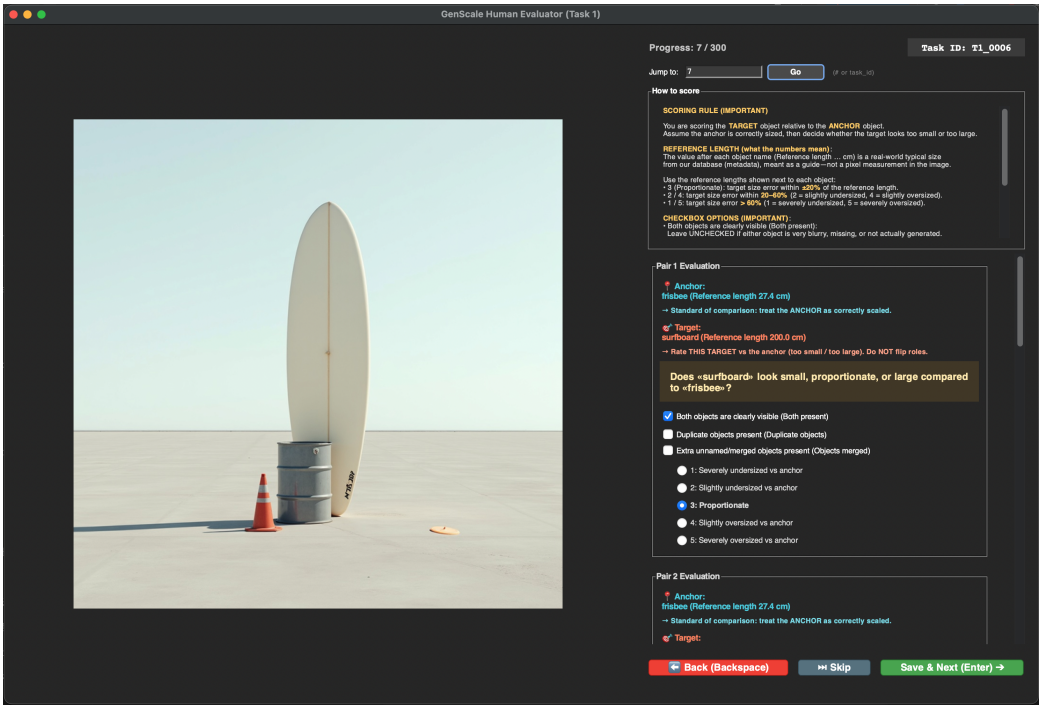

Figure 7: **Human annotation interface.** The interface shows one generated image together with one or more anchor–target pair cards. For each pair, annotators are shown the anchor and target names, their reference physical lengths, pair-specific validity checkboxes, and a five-point ordinal scale for judging whether the target is too small, proportionate, or too large relative to the anchor.

Table 30: **Leave-one-rater-out human reliability.** For each annotator, the reference consensus is rebuilt from the remaining eight annotators only, using modal score aggregation with median-rounded tie breaking. This removes self-inclusion bias from the human reliability estimate. Annotator identities are anonymized and match Tab. 29.

Annotator	Ref. raters	Pairs	Exact (%) ↑	≤ 1 (%) ↑	MAE ↓	QWK ↑	r ↑
Annotator A	8	504	63.69	94.84	0.4246	0.7721	0.7862
Annotator B	8	515	59.61	93.79	0.4757	0.6783	0.6841
Annotator C	8	568	57.04	93.66	0.5000	0.7268	0.7374
Annotator D	8	558	64.70	98.39	0.3692	0.7737	0.7996
Annotator E	8	576	56.25	90.80	0.5347	0.5663	0.5885
Annotator F	8	553	60.76	95.48	0.4412	0.7430	0.7434
Annotator G	8	564	48.23	88.48	0.6365	0.6915	0.7271
Annotator H	8	565	47.26	90.27	0.6248	0.6397	0.6411
Annotator I	8	574	64.46	94.43	0.4146	0.7608	0.7667
Mean $\pm$ std	8	553±26	58.00±6.56	93.35±3.03	0.4913±0.0928	0.7058±0.0695	0.7193±0.0695
Min–Max	8	504–576	47.26–64.70	88.48–98.39	0.3692–0.6365	0.5663–0.7737	0.5885–0.7996

D.4 Gemini Judge Model, Prompt, and Output Schema

For scalable evaluation, we use a Gemini-based VLM judge calibrated against the human consensus. The judge receives the same core evidence as human annotators: the generated image, anchor and target names, reference physical lengths, expected 3D target-to-anchor length ratio, and the five-point ordinal rubric. When the original generation prompt is provided, it is explicitly treated as low-priority disambiguation evidence only. The prompt may help identify intended objects or layout, but it is not allowed to justify a visible scale error simply because the prompt requested accurate scale.

Before pairwise scale scoring, we apply a scene-level validity prefilter to remove images whose visual defects make pairwise scale judgment unreliable. The prefilter checks for duplicate objects, extra unnamed clutter, severe generation artifacts, and whether the intended objects are individually clear. Images failing this prefilter are excluded from scored-pair metrics, while ordinary scale errors remain scored.

The Task 1 judge prompt contains scenario-specific context. For S1 natural-depth cases, the judge is instructed to infer the 3D layout and account for near–far perspective before penalizing a visible size difference. For S2 same-plane cases, the judge is told that the objects are intended to lie on a similar depth plane, so visible size ratios provide stronger evidence unless the image clearly shows large depth separation. The Task 1 output schema is:

```
{
  "reasoning_detection": "...",
  "reasoning_depth_and_perspective": "...",
  "both_objects_present": true,
  "size_score": 3
}
```

The Task 2 judge prompt is specialized for human–product interactions. It treats the human body part as the fixed anchor and the product as the target, while cautioning the judge against over-interpreting cropped hands, close-up faces, partial feet, foreshortened body parts, or product-forward catalog framing as exact metric rulers. The prompt also specifies that bundles, packs, folded products, jewelry displays, and paired products should be judged according to their visible retail presentation rather than a misleading subcomponent. The Task 2 output schema is:

```
{
  "reasoning_detection": "...",
  "reasoning_scale_and_interaction": "...",
  "both_objects_present": true,
  "size_score": 3
}
```

Task 3 does not use a separate scale rubric. Each Task 3 entry stores provenance indicating whether the erroneous source case came from Task 1 or Task 2. For evaluation, Task 3 outputs are routed to the corresponding Task 1 or Task 2 judge template, then remapped back to the Task 3 identifier for aggregation. This ensures that edited images are judged under the same scale criterion as the original failed generation.

D.5 Repeated Sampling and Score Aggregation

For each scorable anchor–target pair, we query the Gemini judge five times. Repeated sampling reduces sensitivity to isolated parsing errors, unstable visual interpretations, or overly literal ratio estimates. Each call returns a numeric score in $\{1, 2, 3, 4, 5\}$ and a pair-validity flag. A pair is included in the final scale metrics only when the required objects are judged present and the scene passes the validity criteria.

Given the five sampled scores for a pair, we first use majority vote. If there is no unique mode, we take the median of the five ordinal scores. This aggregation preserves the discrete ordinal scale while reducing the influence of a single outlier sample. All benchmark models, general-purpose editors, and *Rescale* outputs are evaluated with the same fixed judge, prompt templates, repeated-sampling protocol, and aggregation rule.

D.6 Calibration Diagnostics

We evaluate calibration using exact agreement, within-one agreement, mean absolute error (MAE), Pearson correlation r, and quadratic-weighted kappa (QWK) on the five-point ordinal scale. Exact agreement measures strict label equality, while within-one agreement measures whether two labels differ by at most one ordinal level. MAE is computed as the mean absolute difference between predicted and consensus ordinal scores. QWK is useful because it penalizes large ordinal disagreements more heavily than adjacent disagreements.

The human agreement results in Tabs. 29 and 30 show that the calibration labels are stable but not trivial. Agreement with the aggregate consensus reaches $65.15\% \pm 6.98\%$ exact agreement and $94.03\% \pm 3.16\%$ within-one agreement, while the stricter leave-one-rater-out estimate remains $58.00\% \pm 6.56\%$ exact and $93.35\% \pm 3.03\%$ within-one. This gap is expected because the aggregate-consensus comparison includes the evaluated annotator in the reference, whereas the leave-one-rater-out comparison does not. The leave-one-rater-out QWK of 0.7058 ± 0.0695 indicates that most human disagreements are local on the ordinal scale rather than severe reversals.

Table 31: **Gemini–human alignment with image-level bootstrap confidence intervals.** Gemini-3.1-Pro-preview is evaluated with five repeated samples per pair and majority/median aggregation. Confidence intervals are computed by resampling images rather than individual object pairs.

Split	Images / Pairs	Exact (%) ↑	≤ 1 (%) ↑	MAE ↓	r ↑	QWK ↑
Task 1: S1–S2	185 / 427	61.83 [56.69, 66.96]	96.49 [94.16, 98.35]	0.4169 [0.3565, 0.4786]	0.8473 [0.8108, 0.8774]	0.8373 [0.7968, 0.8696]
Task 2: S3	100 / 100	73.00 [64.00, 81.02]	100.00 [100.00, 100.00]	0.2700 [0.1898, 0.3600]	0.4701 [0.2340, 0.6517]	0.4677 [0.2278, 0.6459]
Task 1+2	285 / 527	63.95 [59.45, 68.45]	97.15 [95.29, 98.72]	0.3890 [0.3373, 0.4416]	0.8328 [0.7970, 0.8636]	0.8234 [0.7839, 0.8563]

The Gemini judge is then compared against the human consensus on the same calibration set. As shown in Tab. 31, Gemini reaches 63.95% exact agreement, 97.15% within-one agreement, MAE 0.3890, Pearson correlation 0.8328, and QWK 0.8234 on the combined Task 1+2 calibration split. These values are at or above the leave-one-rater-out human reliability estimate on the same metrics, supporting the use of the calibrated judge for large-scale model comparison. Task 2 has lower r and QWK despite high exact and within-one agreement because its labels are more concentrated near the plausible-scale score; this makes correlation-based metrics less informative than ordinal error and within-one agreement for that split.

D.7 Bootstrap Confidence Intervals

We report uncertainty for Gemini–human alignment using image-level bootstrap confidence intervals in Tab. 31. The bootstrap resamples images rather than individual object pairs, and all pairwise judgments associated with a sampled image are included together. This preserves the natural clustering induced by multi-pair images and avoids overstating confidence by treating correlated pairs from the same image as independent samples. For each bootstrap replicate, we recompute exact agreement, within-one agreement, MAE, Pearson correlation, and QWK. The reported intervals are percentile confidence intervals over the resulting bootstrap distribution.

E *Rescale* Implementation Details

This appendix expands the implementation of *Rescale*, the model-agnostic scale-correction pipeline used in Sec. 4. We focus on the agentic edit plan, the local insertion interface, the training design of our depth-aware correction backend, and the backend/training ablations. Hardware and compute-resource details are reported separately in the computing-resources appendix and are not repeated here.

E.1 Agent Inputs and Edit-Plan Format

Rescale takes a generated image I together with structured scale metadata. For Task 1, the metadata consists of the prompted object names and their physical reference lengths from the common-object scale knowledge base. For Task 2, it consists of the product reference image, product dimensions, and the human anchor. For Task 3, the input additionally contains an erroneous source image and either a hard auto-discovery prompt (S4) or a precise resize instruction (S5). The inference pipeline is illustrated in Fig. 5.

The agent converts these inputs into a structured local edit plan. For each edit round k, the plan is

$$\pi^{(k)} = \{o_t, o_a, \mathbf{b}_t, \tilde{\mathbf{b}}_t, \alpha, p, q\}, \tag{1}$$

where o_t is the editable target object, o_a is the fixed anchor object, $\mathbf{b}_t$ is the current target bounding box, $\tilde{\mathbf{b}}_t$ is the resized target box, α is the multiplicative resize factor, p is a contact-preserving anchor point, and q is a short natural-language rationale used for verification and debugging. The resized box $\tilde{\mathbf{b}}_t$ is obtained by scaling $\mathbf{b}_t$ by α around p, so that contact points such as object bases, hand-contact regions, or support surfaces remain approximately fixed.

For common-object images, multiple pairwise judgments can implicate the same object. The agent therefore aggregates pairwise evidence into a conservative object-level plan: it edits the object that most consistently explains the observed scale errors and avoids large changes when the pairwise evidence is contradictory. For human–product images, the product is treated as editable and the human body part is fixed. In S5, the target, anchor, correction direction, and resize factor are given directly by the benchmark prompt, so the agent mainly performs grounding and execution. In S4, the agent must first decide whether a correction is needed, choose the target, and infer the correction direction. If the agent cannot localize the objects reliably, finds no clear scale inconsistency, or

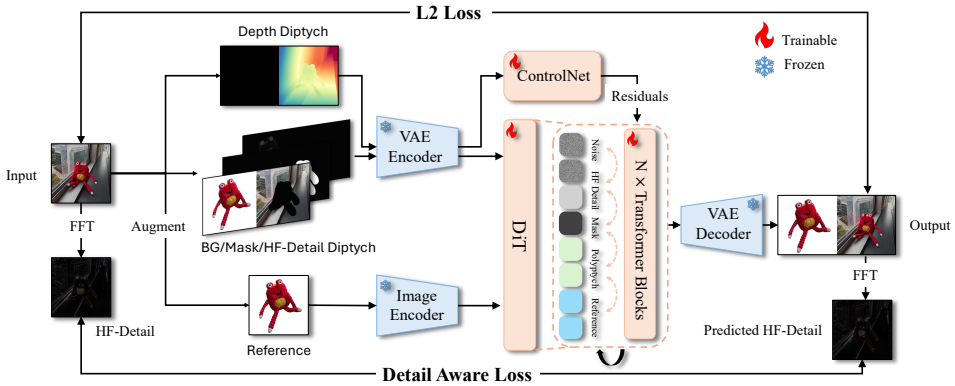

Figure 8: **Training pipeline of our depth-aware correction backend.** The model adopts a diptych-style formulation, with a reference half that provides object identity and appearance, and a target half that contains the completed background and the local region to be corrected. A DiT-based inpainting backbone performs the local reinsertion, a depth ControlNet injects geometry-aware signals from the fused depth condition, and an optional high-frequency branch provides detail cues derived from the reference crop via FFT. Training combines the backbone objective with an optional detail-aware reconstruction loss to improve fidelity after resizing.

predicts that the edit would create severe collisions, truncation, or boundary artifacts, *Rescale* returns a no-edit decision.

E.2 Localization, Segmentation, and Local Editing Interface

Given an edit plan, *Rescale* standardizes all downstream backends to the same local insertion interface. We first refine the target box with multimodal localization and segment the target instance using SAM 2 [52]. The segmented target is extracted before removal and used as the appearance reference R. We then remove the original target from I to obtain a completed background B. This removal step is handled by an off-the-shelf inpainting/editing model, since the purpose of *Rescale* is not to benchmark generic object removal but to test whether a scale-aware local reinsertion can be executed after the source instance has been removed.

The edit mask M is constructed around the resized box $\tilde{\mathbf{b}}_t$ and is deliberately enlarged beyond the expected object support. This gives the insertion backend enough local context for contact shadows, occlusion boundaries, and small background corrections. We also estimate a monocular depth map with DepthAnythingV2 [70]. For depth-aware backends, we fuse the completed-background depth with the resized foreground support to form a correction-aware depth condition D. The resulting backend inputs are therefore

$$(R, B, M, D, \tau), \tag{2}$$

where τ is a compact text instruction describing the target object and the intended resize operation. Backends that do not support depth receive the same reference, background, mask, and text instruction, but ignore D. This interface lets us substitute only the final local generation model while keeping agent planning, localization, segmentation, background completion, and mask construction fixed.

E.3 Depth-Aware Correction Backend

We also explored a specialized local insertion backend tailored to relative-scale correction. As shown in Fig. 8, the model follows a diptych-style in-context formulation: the left half provides a clean reference crop of the object to preserve identity and appearance, while the right half contains the target-side completed background together with the local region to be edited. This design is more suitable for scale correction than a standard inpainting setup, because the model must simultaneously preserve object identity and synthesize a resized insertion that matches the surrounding scene.

Let R denote the clean reference crop, B the completed target background after removing the original instance, I the desired corrected image, M the editable target-side mask, and D the fused depth condition. We construct the training inputs as

$$\mathcal{S} = [R \mid B], \qquad \mathcal{Y} = [R \mid I], \qquad \mathcal{M} = [0 \mid M], \qquad \mathcal{D} = [0 \mid D], \tag{3}$$

where $[\cdot \mid \cdot]$ denotes horizontal concatenation. The left half serves as visual reference, and the right half specifies the local correction problem. During training, B is obtained by removing the ground-truth foreground object from the original image; at inference time, it is replaced by the completed background produced by the upstream removal stage of *Rescale*.

The backbone is a DiT-based inpainting model initialized from the InsertAnything/FLUX-style insertion codebase. The reference crop is encoded by a frozen reference image encoder, while the target half is synthesized by the inpainting transformer. Because relative-scale correction is fundamentally a geometric edit, we add a depth ControlNet branch and feed it the depth diptych $\mathcal{D}$. Its residuals are injected only into the target-half tokens, so that the reference side remains an appearance cue rather than becoming a second geometry target. This design encourages the model to use depth primarily to control the support and extent of the resized object in the target scene.

We also test an optional high-frequency branch for fine-detail preservation. Specifically, we extract an FFT-based high-frequency representation from the reference crop and form

$$\mathcal{H} = [\mathrm{HF}(R) \mid 0], \tag{4}$$

which is projected and injected into the transformer hidden states through a lightweight branch. The motivation is that local resizing can easily smooth textures, edges, and small appearance cues; the high-frequency branch gives the model an explicit signal for recovering such details from the reference object.

A key motivation of this backend is to decouple *editable context extent* from *object extent*. In ordinary mask-conditioned insertion, the model can overfit to the mask boundary and treat it as the intended object boundary. That behavior is undesirable for scale correction, where the mask should reserve enough local context for shadows, contact regions, and boundary adaptation, but the resized object should not necessarily expand to fill the whole editable region. We therefore deliberately perturb and dilate the training masks, and rely on depth as the main cue for the intended object support. Functionally, this makes the backend better aligned with the needs of relative-scale correction, even though standard edited-crop metrics do not directly evaluate this property.

We train only lightweight adaptation modules: LoRA adapters on the DiT backbone, the depth ControlNet branch, and, when enabled, the high-frequency injection branch. The VAE, text encoder, and reference encoder remain frozen. The primary optimization objective is the latent flow-matching objective of the underlying DiT backbone,

$$\mathcal{L}_{\mathrm{fm}} = \|v_\theta(x_t, c) - (x_1 - x_0)\|_2^2, \tag{5}$$

where x_t is the interpolated noisy latent, x_0 is the clean latent, x_1 is Gaussian noise, and c denotes the full set of conditioning inputs. To better preserve local details after resizing, we optionally add a detail-aware reconstruction loss on high-frequency maps,

$$\mathcal{L}_{\mathrm{da}} = \left\| \mathrm{HF}(\hat{I}) \odot M - \mathrm{HF}(I) \odot M \right\|_2^2, \tag{6}$$

and optimize the combined objective

$$\mathcal{L} = \mathcal{L}_{\mathrm{fm}} + \lambda_{\mathrm{da}} \mathcal{L}_{\mathrm{da}}. \tag{7}$$

Training data construction. We construct correction-style training tuples from the same family of segmentation, video-object, saliency, fashion, and insertion datasets used by the insertion backbone, including SAM, LVIS, saliency datasets, YouTubeVOS, VIPSeg, MOSE, VITON-HD, and AnyInsertion. Each training sample is converted into a reference–target diptych by extracting a foreground reference crop, removing the original instance from the target side, and using the original image as supervision. We further apply appearance and geometric augmentations to the reference crop, synthetic partial occlusion to make the reference less idealized, and mask perturbation/dilation so that the model cannot trivially infer object extent from mask shape alone. These choices make the training task closer to the actual scale-correction setting, where the reference may be incomplete and the editable region must include both the resized object and its surrounding context.

E.4 Insertion Backend Substitution Study

Tab. 32 evaluates the final local generation step while holding the rest of *Rescale* fixed. For each input, we reuse the same agent plan, target box, segmentation mask, completed background, reference crop, and depth estimate, and replace only the insertion backend. This isolates whether differences come from the local synthesis model rather than from upstream diagnosis or preprocessing.

Table 32: **Insertion backend substitution under fixed *Rescale* plans.** We keep agent planning, localization, segmentation, background completion, mask construction, and depth estimation fixed, and replace only the final local generation backend. Metrics are computed on Task 1 edited-object bounding boxes; higher is better.

	Visual Consistency				Generation Quality	
Insertion backend	**CLIP-I (%) ↑**	**DINO (%) ↑**	**SSIM (%) ↑**	**SSIM-HF (%) ↑**	**LAION-Aes ↑**	**Q-Align-IQ ↑**
Ours (depth-aware)	86.5	70.8	**50.2**	**75.6**	3.68	3.58
InsertAnything [57]	**87.7**	**75.6**	48.8	75.0	**3.71**	**3.61**
CreatiLayout [73]	75.9	38.3	31.4	70.7	3.30	2.46
Bifrost [35]	82.5	66.2	40.3	73.5	3.65	3.44

Table 33: **Training-design ablation for the depth-aware correction backend.** Each row incrementally adds a component to the same base diptych architecture: FLUX-style guidance conditioning, depth ControlNet conditioning, and the detail-aware loss plus high-frequency branch. Metrics are computed on Task 1 edited-object bounding boxes; higher is better.

	Visual Consistency				Generation Quality	
Training configuration	**CLIP-I (%) ↑**	**DINO (%) ↑**	**SSIM (%) ↑**	**SSIM-HF (%) ↑**	**LAION-Aes ↑**	**Q-Align-IQ ↑**
Baseline	74.2	30.6	44.1	**75.8**	3.19	1.91
Baseline + guidance	77.6	44.8	40.8	74.2	3.40	2.13
Baseline + guidance + depth	**79.4**	47.1	45.0	75.6	**3.58**	**2.19**
Baseline + guidance + depth + DAL + HF	**79.4**	**47.3**	**45.3**	**75.8**	3.57	**2.19**

The original InsertAnything checkpoint obtains the best CLIP-I, DINO, LAION-Aes, and Q-Align-IQ scores, while our depth-aware backend is slightly better on SSIM and SSIM-HF. The two are therefore broadly comparable under these generic edited-object crop metrics, but our specialized backend does not outperform the stronger codebase checkpoint overall. CreatiLayout performs substantially worse on visual consistency, which is expected because it is driven by text and boxes rather than a reference image, so it cannot reliably preserve the edited object's appearance. Bifrost is competitive but remains below InsertAnything and our tuned backend on most identity-preservation metrics.

This result should be interpreted with an important caveat. The metrics in Tab. 32 measure visual consistency and no-reference quality inside edited boxes; they do not measure whether the backend can use a large editable mask without forcing the object to fill the mask. Our depth-aware backend was designed for precisely that functional requirement. Thus, although the current automatic metrics favor the original InsertAnything checkpoint slightly, the specialized backend remains a useful design exploration for scale-aware insertion. Developing an evaluation protocol that directly measures mask–extent decoupling and depth-controlled resizing is left for future work.

E.5 Correction-Model Training Ablation

Tab. 33 ablates the design choices of our specialized correction backend. The baseline uses the same diptych insertion formulation without the additional guidance, depth, detail-aware loss, or high-frequency branch. Adding FLUX-style guidance substantially improves representation-level consistency, especially DINO, and also improves both no-reference quality metrics. Adding depth further improves CLIP-I, DINO, SSIM, LAION-Aes, and Q-Align-IQ, supporting the claim that explicit geometry is useful for scale-aware local insertion. Adding the detail-aware loss and high-frequency branch gives only marginal additional gains on these aggregate metrics: DINO, SSIM, and SSIM-HF increase slightly, while LAION-Aes and Q-Align-IQ are essentially unchanged.

Overall, the ablation suggests that guidance and depth conditioning are the main effective components for this backend, whereas the detail-preservation branch is not strongly reflected by the current crop-level metrics. Combined with the backend substitution study, these results indicate that our depth-aware model is a functional attempt to make local insertion more appropriate for scale correction, but the generic InsertAnything checkpoint remains a very strong synthesis backend. Future work should train a specialized model with losses and evaluation metrics that directly target numeric resize fidelity, contact preservation, and the ability to decouple mask extent from object extent.

F Full Quantitative Results

F.1 Metrics and Statistical Protocol

All quantitative results use the calibrated Gemini judge described in Sec. D. The atomic unit is an anchor–target pair rather than an image. For each valid pair, the judge returns an ordinal scale score $s \in \{1, 2, 3, 4, 5\}$, where $s = 3$ denotes a physically plausible target size relative to the anchor, $s < 3$ denotes an undersized target, and $s > 3$ denotes an oversized target. We report scale error as $|s - 3|$ averaged over scored pairs. Plausible is the fraction of pairs with $s = 3$; Too small and Too large are the fractions with $s \in \{1, 2\}$ and $s \in \{4, 5\}$, respectively; Severe is the fraction with $s \in \{1, 5\}$.

All tables are computed after the same scene-level validity prefilter used in the main experiments. Valid img. / pairs denotes the number of images with at least one scored pair and the number of scored pairs after filtering. Bracketed ranges denote 95% confidence intervals from 2,000 bootstrap resamples. For one-pass generation or editing results, CIs are reported for pair-level mean scale error. For correction experiments, matched-pair tables only include pairs that are scored both before and after editing; CIs are reported for before/after mean scale error and Gain. Gain is the reduction in matched-pair scale error, so positive values indicate improvement. B / W counts pairs whose absolute error $|s - 3|$ becomes smaller or larger after editing. Directional rate columns are empirical percentages and are kept as point estimates to avoid making the wide appendix tables unreadable.

F.2 Task 1 Full Results

Tab. 34 expands the main Task 1 table with valid coverage and the full error-direction breakdown. MR denotes mean-regression error: cases where the physical target is smaller than the anchor but judged too large, or physically larger than the anchor but judged too small.

F.3 Task 2 Full Results

Tab. 35 expands the main Task 2 table with the directional error rates. Because Task 2 contains one human–product relation per image, image-level and pair-level coverage are nearly identical after filtering.

F.4 Task 3 Full Results

Tab. 36 reports the full Task 3 results for general-purpose editors. S4 requires autonomous scale-error discovery, while S5 gives the target object, reference object, resize direction, and scale factor. The before-edit row is the erroneous source image set used to construct Task 3.

F.5 *Rescale* Correction on Task 1

Tab. 37 reports matched-pair before–after results for applying *Rescale* to Task 1 generations from each source model. The table uses only pairs scored on both the original and corrected image.

F.6 *Rescale* Correction on Task 2

Tab. 38 reports matched-pair *Rescale* correction results for Task 2. Since each Task 2 image has one evaluated pair, matched pairs are equivalent to matched images after filtering.

F.7 *Rescale* Correction on Task 3

Tab. 39 reports *Rescale* on the same Task 3 correction benchmark used for the general-purpose editors in Tab. 36. Both S4 and S5 use the released Task 3 source images; S5 additionally provides the explicit target, anchor, direction, and scale factor.

F.8 Identity Preservation and Visual-Quality Metrics

Tab. 40 reports visual consistency and no-reference quality metrics for *Rescale* corrections. CLIP-I and DINO measure reference consistency between the original and corrected images. SSIM and SSIM-HF measure image-level and high-frequency preservation. LAION-Aes and Q-Align-IQ measure no-reference image quality before and after correction.

Table 34: **Task 1 full results.** Results are split by S1, S2, and all Task 1 entries. Error is mean $|s-3|$ with 95% CI in brackets. Plausible denotes score 3; Too small denotes scores 1–2; Too large denotes scores 4–5; Severe denotes scores 1 or 5. MR denotes mean-regression error.

Model	Split	Valid img. / pairs	Error ↓	Plaus. (%) ↑	Too small (%) ↓	Too large (%) ↓	Severe (%) ↓	MR (%) ↓
Nano Banana 2	S1	199 / 564	0.505 [0.449,0.566]	65.1	17.0	17.9	15.6	15.8
	S2	196 / 557	0.691 [0.630,0.750]	47.8	25.1	27.1	16.9	47.9
	All	395 / 1121	0.598 [0.553,0.644]	56.5	21.1	22.5	16.2	31.8
GPT-Image-2	S1	199 / 564	0.784 [0.716,0.856]	51.1	20.7	28.2	29.4	12.1
	S2	200 / 572	0.629 [0.570,0.691]	52.8	22.7	24.5	15.7	35.8
	All	399 / 1136	0.706 [0.659,0.753]	51.9	21.7	26.3	22.5	24.0
Z-Image-Turbo	S1	193 / 534	0.644 [0.581,0.708]	53.2	22.8	24.0	17.6	35.8
	S2	181 / 470	0.881 [0.806,0.955]	38.1	33.0	28.9	26.2	55.7
	All	374 / 1004	0.755 [0.705,0.803]	46.1	27.6	26.3	21.6	45.1
Grok Image	S1	198 / 563	0.599 [0.533,0.666]	58.8	19.9	21.3	18.7	26.8
	S2	200 / 573	0.979 [0.911,1.047]	34.7	31.1	34.2	32.6	61.1
	All	398 / 1136	0.790 [0.742,0.838]	46.7	25.5	27.8	25.7	44.1
Qwen-Image 2512	S1	196 / 542	0.699 [0.629,0.771]	53.3	22.5	24.2	23.2	37.3
	S2	199 / 563	1.103 [1.039,1.171]	27.2	35.7	37.1	37.5	67.5
	All	395 / 1105	0.905 [0.855,0.956]	40.0	29.2	30.8	30.5	52.7
FLUX.2	S1	176 / 453	0.837 [0.764,0.916]	45.0	27.2	27.8	28.7	40.6
	S2	178 / 490	1.212 [1.141,1.284]	23.3	38.8	38.0	44.5	69.4
	All	354 / 943	1.032 [0.978,1.085]	33.7	33.2	33.1	36.9	55.6
SD3.5-Large	S1	180 / 503	0.960 [0.887,1.030]	36.6	32.6	30.8	32.6	46.7
	S2	166 / 446	1.184 [1.110,1.260]	24.9	36.3	38.8	43.3	69.7
	All	346 / 949	1.065 [1.013,1.118]	31.1	34.4	34.6	37.6	57.5

Table 35: **Task 2 full results.** Each Task 2 image contains one product–human scale relation. Error is mean $|s-3|$ with 95% CI in brackets. The remaining columns report the full score-direction breakdown for S3.

Model	Valid img. / pairs	Error ↓	Plaus. (%) ↑	Too small (%) ↓	Too large (%) ↓	Severe (%) ↓
GPT-Image-2	294 / 294	0.231 [0.180,0.289]	78.9	3.4	17.7	2.0
Nano Banana 2	295 / 295	0.268 [0.210,0.325]	75.6	4.4	20.0	2.4
Seedream v4.5	295 / 295	0.302 [0.237,0.366]	74.9	8.1	16.9	5.1
Qwen-Image-Edit-2511	297 / 297	0.327 [0.266,0.391]	71.0	4.4	24.6	3.7
FLUX.1 Kontext-dev	291 / 291	0.423 [0.354,0.491]	63.2	15.1	21.6	5.5
SD3.5-Large + IP-Adapter	270 / 270	0.600 [0.515,0.685]	52.6	7.8	39.6	12.6

G Additional Qualitative Results

This section provides additional qualitative examples for the three benchmark tasks and for *Rescale* correction. The examples are intended to visualize the input conditions, model outputs, and typical correction behavior; quantitative conclusions are based on the calibrated metrics reported in the main paper and appendix tables.

Table 36: **Task 3 full results for general-purpose editors.** Error and score-direction rates are computed on each editor's valid scored outputs. Error and Gain include 95% CIs in brackets. Gain and B / W are computed on matched before–after pairs relative to the corresponding before-edit source image. SD3.5-Large + IP-Adapter has substantially lower scored coverage and should be interpreted cautiously.

Model	Split	Valid img. / pairs	Error ↓	Plaus. (%) ↑	Too small (%) ↓	Too large (%) ↓	Severe (%) ↓	Gain ↑	B / W
Before edit	S4	98 / 98	1.265 [1.112,1.418]	20.4	31.6	48.0	46.9	–	–
	S5	98 / 98	1.265 [1.112,1.418]	20.4	31.6	48.0	46.9	–	–
	All	196 / 196	1.265 [1.158,1.372]	20.4	31.6	48.0	46.9	–	–
GPT-Image-2	S4	97 / 97	0.959 [0.784,1.134]	40.2	28.9	30.9	36.1	+0.333 [0.198,0.469]	31 / 4
	S5	98 / 98	0.408 [0.296,0.531]	66.3	16.3	17.3	7.1	+0.866 [0.691,1.031]	60 / 5
	All	195 / 195	0.682 [0.569,0.795]	53.3	22.6	24.1	21.5	+0.601 [0.487,0.720]	91 / 9
Nano Banana 2	S4	99 / 99	1.030 [0.879,1.182]	31.3	27.3	41.4	34.3	+0.237 [0.103,0.381]	24 / 7
	S5	98 / 98	0.929 [0.776,1.092]	33.7	25.5	40.8	26.5	+0.340 [0.206,0.474]	31 / 4
	All	197 / 197	0.980 [0.873,1.091]	32.5	26.4	41.1	30.5	+0.289 [0.196,0.381]	55 / 11
FLUX.1 Kontext-dev	S4	99 / 99	1.293 [1.131,1.435]	21.2	30.3	48.5	50.5	-0.021 [-0.113,0.072]	8 / 9
	S5	94 / 94	0.957 [0.798,1.128]	36.2	25.5	38.3	31.9	+0.280 [0.140,0.419]	27 / 7
	All	193 / 193	1.130 [1.010,1.249]	28.5	28.0	43.5	41.5	+0.126 [0.042,0.211]	35 / 16
Qwen-Image-Edit-2511	S4	99 / 99	1.253 [1.091,1.414]	23.2	29.3	47.5	48.5	+0.020 [-0.071,0.112]	9 / 4
	S5	91 / 91	1.000 [0.824,1.165]	34.1	25.3	40.7	34.1	+0.253 [0.099,0.407]	23 / 5
	All	190 / 190	1.132 [1.016,1.247]	28.4	27.4	44.2	41.6	+0.132 [0.048,0.222]	32 / 9
Seedream v4.5	S4	100 / 100	1.170 [1.010,1.330]	28.0	37.0	35.0	45.0	+0.102 [-0.031,0.224]	18 / 8
	S5	96 / 96	1.104 [0.938,1.281]	35.4	31.2	33.3	45.8	+0.200 [0.032,0.368]	25 / 9
	All	196 / 196	1.138 [1.020,1.260]	31.6	34.2	34.2	45.4	+0.150 [0.047,0.254]	43 / 17
SD3.5-Large + IP-Adapter	S4	15 / 15	0.933 [0.467,1.333]	40.0	6.7	53.3	33.3	-0.200 [-0.667,0.268]	3 / 6
	S5	59 / 59	1.305 [1.102,1.508]	22.0	44.1	33.9	52.5	-0.158 [-0.368,0.053]	7 / 13
	All	74 / 74	1.230 [1.041,1.405]	25.7	36.5	37.8	48.6	-0.167 [-0.361,0.028]	10 / 19

G.1 Benchmark Record Examples

To make the benchmark format explicit, we show one randomly sampled entry from each scenario in Figs. 9 to 13. For readability, absolute local path prefixes are shortened to dataset-relative or generated-image-relative paths, while the benchmark fields are otherwise preserved.

G.2 Task 1: Common-Object Generation Examples

Figs. 14 and 15 show additional Task 1 generations across the evaluated text-to-image models. S1 permits natural depth variation, while S2 constrains objects to approximately the same plane, making relative-size errors more visually exposed.

G.3 Task 2: Human-Product Generation Examples

Fig. 16 shows additional Task 2 examples. The leftmost column shows the product reference image, and the remaining columns show image-conditioned generations from different models. These examples illustrate that preserving product identity and realizing metric scale relative to a human anchor are distinct requirements.

G.4 Task 3: General-Purpose Editor Correction Examples

Fig. 17 shows qualitative results for general-purpose editors on Task 3. The first column is the erroneous source image, and the remaining columns are editor outputs under the corresponding correction prompt. The examples illustrate the diagnosis–execution gap: editors often preserve visual realism but may under-correct, over-correct, or regenerate content beyond the intended localized scale edit.

Table 37: ***Rescale* correction on Task 1.** Metrics are computed on matched scorable pairs before and after correction. Error before, Error after, and Gain include 95% CIs in brackets. Plaus. and Severe are reported as before / after percentages.

Source model	Split	Matched pairs	Error before ↓	Error after ↓	Gain ↑	Plaus. before / after (%) ↑	Severe before / after (%) ↓	B / W
Nano Banana 2	S1	536	0.494 [0.431,0.558]	0.388 [0.332,0.448]	+0.106 [0.043,0.174]	65.7 / 73.9	15.1 / 12.7	107 / 61
	S2	544	0.688 [0.625,0.752]	0.379 [0.327,0.432]	+0.309 [0.241,0.375]	48.2 / 69.3	16.9 / 7.2	188 / 58
	All	1080	0.592 [0.546,0.636]	0.383 [0.345,0.424]	+0.208 [0.162,0.256]	56.9 / 71.6	16.0 / 9.9	295 / 119
GPT-Image-2	S1	508	0.768 [0.701,0.843]	0.476 [0.413,0.541]	+0.291 [0.217,0.364]	52.6 / 68.3	29.3 / 15.9	141 / 44
	S2	556	0.622 [0.563,0.683]	0.417 [0.365,0.471]	+0.205 [0.138,0.272]	53.4 / 66.9	15.6 / 8.6	158 / 70
	All	1064	0.692 [0.644,0.743]	0.445 [0.404,0.491]	+0.246 [0.195,0.297]	53.0 / 67.6	22.2 / 12.1	299 / 114
Z-Image-Turbo	S1	497	0.644 [0.577,0.708]	0.370 [0.314,0.425]	+0.274 [0.201,0.348]	53.3 / 71.8	17.7 / 8.9	161 / 54
	S2	441	0.889 [0.816,0.964]	0.499 [0.435,0.562]	+0.390 [0.304,0.483]	37.9 / 61.0	26.8 / 10.9	184 / 59
	All	938	0.759 [0.711,0.811]	0.431 [0.389,0.473]	+0.328 [0.272,0.386]	46.1 / 66.7	22.0 / 9.8	345 / 113
Grok Image	S1	531	0.589 [0.525,0.657]	0.339 [0.284,0.394]	+0.250 [0.181,0.324]	59.5 / 75.5	18.5 / 9.4	148 / 55
	S2	567	0.968 [0.899,1.034]	0.554 [0.499,0.616]	+0.414 [0.337,0.492]	35.1 / 58.0	31.9 / 13.4	241 / 66
	All	1098	0.785 [0.736,0.832]	0.450 [0.407,0.489]	+0.335 [0.281,0.390]	46.9 / 66.5	25.4 / 11.5	389 / 121
Qwen-Image 2512	S1	490	0.704 [0.631,0.773]	0.461 [0.400,0.529]	+0.243 [0.163,0.324]	53.3 / 68.4	23.7 / 14.5	152 / 69
	S2	544	1.107 [1.035,1.173]	0.388 [0.340,0.438]	+0.719 [0.642,0.792]	27.0 / 66.5	37.7 / 5.3	315 / 40
	All	1034	0.916 [0.867,0.966]	0.423 [0.382,0.463]	+0.493 [0.435,0.552]	39.5 / 67.4	31.0 / 9.7	467 / 109
FLUX.2	S1	409	0.819 [0.741,0.902]	0.435 [0.367,0.504]	+0.384 [0.301,0.474]	46.5 / 69.2	28.4 / 12.7	150 / 41
	S2	439	1.216 [1.141,1.289]	0.617 [0.549,0.688]	+0.599 [0.510,0.686]	23.2 / 54.4	44.9 / 16.2	224 / 40
	All	848	1.025 [0.967,1.083]	0.529 [0.481,0.581]	+0.495 [0.429,0.554]	34.4 / 61.6	36.9 / 14.5	374 / 81
SD3.5-Large	S1	461	0.939 [0.868,1.017]	0.618 [0.549,0.685]	+0.321 [0.232,0.408]	38.0 / 57.7	31.9 / 19.5	174 / 71
	S2	375	1.165 [1.085,1.245]	0.656 [0.576,0.739]	+0.509 [0.408,0.605]	25.6 / 54.4	42.1 / 20.0	168 / 40
	All	836	1.041 [0.983,1.097]	0.635 [0.580,0.690]	+0.406 [0.337,0.468]	32.4 / 56.2	36.5 / 19.7	342 / 111

Table 38: ***Rescale* correction on Task 2.** Metrics are computed on matched scorable pairs before and after correction. Error before, Error after, and Gain include 95% CIs in brackets. Plaus. and Severe are reported as before / after percentages.

Source model	Matched pairs	Error before ↓	Error after ↓	Gain ↑	Plaus. before / after (%) ↑	Severe before / after (%) ↓	B / W
GPT-Image-2	290	0.231 [0.176,0.286]	0.090 [0.055,0.128]	+0.141 [0.100,0.186]	79.0 / 92.1	2.1 / 1.0	40 / 1
Nano Banana 2	291	0.261 [0.210,0.320]	0.086 [0.052,0.120]	+0.175 [0.124,0.227]	76.3 / 92.4	2.4 / 1.0	54 / 7
Seedream v4.5	285	0.295 [0.228,0.361]	0.168 [0.119,0.221]	+0.126 [0.077,0.179]	75.8 / 86.3	5.3 / 3.2	40 / 9
Qwen-Image-Edit-2511	284	0.306 [0.246,0.373]	0.144 [0.099,0.194]	+0.162 [0.102,0.222]	73.2 / 87.7	3.9 / 2.1	51 / 12
FLUX.1 Kontext-dev	281	0.406 [0.338,0.477]	0.228 [0.171,0.292]	+0.178 [0.121,0.235]	65.1 / 81.5	5.7 / 4.3	53 / 9
SD3.5-Large + IP-Adapter	246	0.565 [0.480,0.655]	0.256 [0.191,0.321]	+0.309 [0.240,0.382]	54.5 / 78.5	11.0 / 4.1	69 / 4

G.5 *Rescale* Correction Examples

Figs. 18 to 22 show additional before–after examples from *Rescale*. The examples emphasize localized correction: the target object is resized and reinserted while the surrounding scene, reference object, lighting, and background are intended to remain fixed.

G.6 Failure Cases and Limitations

Figs. 23 and 24 show representative failure cases. In S4, failures often arise from the upstream diagnosis problem: the agent may choose an overly conservative correction, select the wrong object, or estimate an inaccurate resize factor. In S5, where the edit plan is already specified, failures more often reflect insertion-backend limitations, including imperfect boundary blending, shape distortion after resizing, or unintended local content changes. These cases indicate that relative-scale correction requires a backend that can decouple the edit mask from the final object size and shape while preserving object identity.

Table 39: ***Rescale* correction on Task 3.** Metrics are computed on matched scorable pairs before and after correction. Error before, Error after, and Gain include 95% CIs in brackets. Plaus. and Severe are reported as before / after percentages.

Split	Matched pairs	Error before ↓	Error after ↓	Gain ↑	Plaus. before / after (%) ↑	Severe before / after (%) ↓	B / W
S4	93	1.258 [1.097,1.409]	0.548 [0.409,0.710]	+0.710 [0.495,0.914]	21.5 / 58.1	47.3 / 12.9	51 / 8
S5	93	1.258 [1.097,1.409]	0.548 [0.409,0.710]	+0.710 [0.495,0.914]	21.5 / 58.1	47.3 / 12.9	51 / 8
All	186	1.258 [1.145,1.366]	0.548 [0.446,0.645]	+0.710 [0.559,0.855]	21.5 / 58.1	47.3 / 12.9	102 / 16

Table 40: **Identity preservation and visual quality after *Rescale* correction.** Higher values indicate better preservation or quality. Percentages for LAION-Aes and Q-Align-IQ denote relative change after correction.

Setting	CLIP-I (%) ↑	DINO (%) ↑	SSIM (%) ↑	SSIM-HF (%) ↑	LAION-Aes before / after ↑	Q-Align-IQ before / after ↑
Task 1	94.8	90.2	88.8	92.4	5.83 / 5.73 (−1.7%)	4.74 / 4.67 (−1.5%)
Task 2	92.4	84.5	73.5	82.1	4.99 / 4.96 (−0.8%)	4.88 / 4.88 (0.0%)
Task 3	95.6	88.9	89.3	92.9	5.83 / 5.73 (−1.2%)	4.76 / 4.67 (−1.9%)

H Comparison to GenSpace

GenSpace is a benchmark that runs parallel to our work; here we analyze where exactly GenScale offers novelty over GenSpace. Specifically, GenSpace subdivides spatial awareness into a number of individual tests, each examining a particular capability of text to image models. Of these tests, Relative Size determines that a model generated objects correctly with respect to relative sizing if the larger object, as specified in the prompt, has at least 1.2 times the predicted volume of the smaller object. In contrast, GenScale uses ground-truth size ratios specific to each object pair, which we expect to make it a more fine-grained evaluator of relative size. To verify this, we score each object pair in Task 1 on GenSpace's Relative Size test, and compute the agreement between GenSpace and our benchmark. We drop the object pairs where GenSpace failed to detech one or more of the named objects (only 14.5 % of pairs), and Task 2 as these are OOD for GenSpace and thus would not serve as a fair comparison. The results of this experiment are shown in Table 41, and given the increased detection of sizing errors by GenScale, we conclude that it is the more fine-grained benchmark.

Table 41: **Comparison between GenScale and GenSpace.** These percentages are calculated over 5,357 object pairs from Task 1.

		GenSpace	
		Correct	Incorrect
Ours	Correct	25.2	4.1
	Incorrect	60.1	10.6

I Release, Licenses, Safeguards, and Societal Considerations

I.1 Benchmark and Code Release

We will release the benchmark metadata, evaluation prompts, human-annotation interface materials, judge prompts, and evaluation scripts to support reproducible comparison on relative-scale generation and editing. The released benchmark metadata follows the schema described in Sec. C: each instance records the task/scenario identifier, generation or editing prompt, object metadata, pair-level physical-size information, expected relative-scale interval, and the fields needed to reconstruct the automatic and human-calibrated evaluation protocol. For Task 2, the release includes the product reference information needed for evaluation, subject to the license terms of the original source assets. For Task 3, we release the erroneous-image identifiers, correction instructions, target–anchor pair metadata, and before/after evaluation records.

We will also release the code used for benchmark construction, pairwise score aggregation, bootstrap confidence intervals, calibration diagnostics, and *Rescale* inference. The code release will exclude API keys, private credentials, and closed-source model weights. For proprietary generation and

editing models, we provide prompts, wrappers, and evaluation scripts rather than redistributing model weights. Generated images and third-party assets will only be redistributed when allowed by the corresponding provider or dataset license; otherwise, we release identifiers, prompts, metadata, and scripts sufficient to reproduce the evaluation under the user's own access to those resources.

I.2 Existing Assets and Licenses

GenScale builds on existing public assets and pretrained tools, and the release will preserve their attribution and license requirements. Task 2 uses product reference images and product metadata from the Amazon Berkeley Objects dataset; all ABO-derived fields and images are released only under the applicable upstream license and attribution requirements. The benchmark metadata records source identifiers so that downstream users can trace each reference image or product record back to its original dataset entry. For all pretrained components used in detection, segmentation, depth estimation, evaluation, or insertion, including open-source perception models and insertion backbones, we use the publicly released checkpoints under their respective licenses and do not claim ownership of those assets.

For generated outputs from commercial or closed-source models, the release follows the terms of the corresponding service providers. If redistribution of an output image is not permitted, we release the prompt, task metadata, and evaluation record rather than the image itself. All release files will include a license summary and citation list specifying which components are original to this work and which are inherited from external assets.

I.3 Responsible Release Safeguards

The benchmark is designed to evaluate physical-scale consistency rather than identity, demographic attributes, or private personal information. Prompts avoid requesting named individuals, sensitive personal attributes, or targeted depictions of real people. Human figures in Task 2 are used as generic scale anchors for product-size evaluation, and the evaluation focuses only on the relative size between the product and the human anchor.

Before release, we remove or withhold samples that contain unsafe, private, or personally identifying content, and we retain the invalid-pair filtering used in the evaluation pipeline. We also avoid releasing private API credentials, hidden model outputs, or any data that would enable unauthorized use of proprietary systems. Because relative-scale correction can make synthetic images more physically plausible, we frame *Rescale* as a benchmarked research tool for diagnosing and correcting geometric scale errors rather than as a general-purpose tool for deceptive image manipulation. Users of the released code and benchmark will be asked to follow the licenses of the underlying datasets and model providers and to disclose synthetic-image editing when outputs are used outside research evaluation.

I.4 Human Annotation Compensation and Participant Risk

Human annotation was used to calibrate the ordinal VLM judge and to validate the evaluation protocol. Annotators were asked to inspect generated images and assign pairwise relative-scale labels according to the provided rubric. The task involved ordinary visual judgments over generated or product-reference images and did not require annotators to disclose demographic attributes, private personal information, political views, health information, or other sensitive data.

Each annotation session was designed to take approximately one to two hours. Annotators received a fixed $30 gift card, corresponding to an effective hourly compensation of $15–$30 depending on completion time. Using the conservative two-hour estimate, this compensation meets or exceeds the applicable minimum-wage requirement for the annotation setting. Annotators were compensated for completing the task rather than for matching any model prediction or preferred answer, reducing incentive pressure to provide a particular label. The task posed minimal participant risk: the images were non-sensitive visual scenes, the instructions did not ask annotators to make personal disclosures, and the collected annotations were used only in aggregate for evaluator calibration.

I.5 Broader Impacts

Reliable relative-scale evaluation can improve the physical realism of image generation and editing systems. This is useful for scientific benchmarking, product visualization, assistive design tools, embodied-AI simulation, and other settings where object size should be physically plausible. By

making scale errors measurable, *GenScale* can also help reveal systematic weaknesses in current generative models that are not captured by generic image-quality or text-alignment metrics.

At the same time, improved scale correction may make synthetic images harder to recognize as edited, especially in contexts where unrealistic object size would otherwise reveal manipulation. This risk is not unique to our method, but *Rescale* can reduce one class of visual artifact. We therefore restrict our release to research-oriented benchmark code and metadata, preserve source and edit provenance where possible, and encourage downstream users to disclose image editing in public-facing applications. The benchmark should not be interpreted as a detector of authenticity or as evidence that a generated image is physically correct in all respects; it evaluates a specific pairwise relative-scale criterion.

I.6 Additional Limitations

Several limitations remain beyond the task-specific analyses in the main paper. First, *GenScale* evaluates pairwise relative scale rather than full 3D scene geometry. A generated image may receive a plausible pairwise score while still containing other geometric, physical, or semantic errors. Second, the ordinal rubric intentionally abstracts away from exact pixel measurements, which improves robustness to perspective and occlusion but cannot provide a complete metric reconstruction of object size. Third, Task 2 depends on the coverage and accuracy of product metadata, and product categories with ambiguous dimensions, deformable shapes, or unusual usage contexts remain challenging.

The automatic evaluator is calibrated against human consensus, but it can still inherit biases from the VLM judge and from the sampled human annotations. The calibration set is intended to verify evaluator reliability for the benchmark distribution, not to prove universal agreement across all possible object categories, camera viewpoints, or artistic styles. Similarly, *Rescale* is a post-generation correction pipeline rather than a fully 3D-aware scene re-renderer. It can resize and reinsert localized objects while preserving much of the surrounding scene, but it may fail under severe occlusion, complex contact geometry, strong shadows or reflections, transparent objects, articulated objects, or cases where correcting scale requires global scene re-composition. These limitations motivate future work on physically grounded generative models that jointly reason about object identity, metric size, depth, support relations, and scene layout.

Field	Value
task_id	T1_0082
scenario	S1_Natural_Depth
num_objects	2
objects_included	computer mouse; towel
prompt	Strictly accurate real-world physical proportions, natural depth and perspective. A tiny computer mouse rests slightly in front of a large, folded bath towel, emphasizing their size contrast. They sit isolated on the smooth concrete floor of a vast, empty photography studio, casting natural shadows in clear depth. Photorealistic.
gt_ratios	computer mouse_to_towel: target ratio 0.084; acceptable range [0.081, 0.087].
reference_image_path	null

Figure 9: **Example benchmark record for S1: Natural Depth.** Task 1 S1 entries contain only a text prompt and pairwise physical-ratio metadata. No metric object dimensions or reference image are provided to the generator.

Field	Value
task_id	T1_0238
scenario	S2_Extreme_Contrast
num_objects	3
objects_included	violin; billiard ball; bowl
prompt	A wooden violin, a polished billiard ball, and a ceramic bowl rest side-by-side on the endless grey floor of a massive, empty concrete warehouse. The sharp, photorealistic lighting emphasizes the textural contrast between the wood, resin, and ceramic against the stark, neutral background. Photorealistic, objects placed on the exact same depth plane, strictly accurate real-world physical proportions.
gt_ratios	violin_to_billiard ball: target ratio 10.526; acceptable range [10.120, 10.948]. violin_to_bowl: target ratio 4.000; acceptable range [3.843, 4.163]. billiard ball_to_bowl: target ratio 0.380; acceptable range [0.365, 0.395].
reference_image_path	null

Figure 10: **Example benchmark record for S2: Same Plane.** S2 uses the same common-object metadata as S1, but prompts constrain objects to a shared depth plane so that image-space size differences more directly expose real-world scale errors.

Field	Value
task_id	T2_0202
scenario	S3_Human_Product_Anchor
num_objects	2
objects_included	trash can; human foot/leg
prompt	A photorealistic, premium catalog shot of a rectangular steel slim trash can. The bin stands about 64.5 centimeters tall, reaching just below the knee of a person standing next to it, with a length of 46.5 centimeters and a width of 29.5 centimeters. A human foot wearing a casual sneaker is naturally pressing down on the step pedal at the base of the bin to open the lid, while the lower leg is visible beside it to demonstrate the scale. The camera captures the scene from a mild three-quarter perspective with bright, even studio lighting. The sleek metallic silhouette of the trash can remains entirely unobstructed to clearly display its real-world bulk and aspect ratio. Strict real-world physical proportions between the human foot, leg, and the product.
gt_ratios	trash can_to_human foot/leg: target ratio 2.434; acceptable range [2.271, 2.613].
reference_image_path	ABO/images/original/6b/6bca188f.jpg
raw_listing_title	Amazon Basics Rectangular Soft Close Steel Slim Trash Can 40
refine_meta	refined with Gemini; model gemini-3.1-pro-preview.
product_scale	source: ABO Dataset; category id: B07PCXZ14V; length 46.51 cm; width 29.49 cm; height 64.49 cm; typical length 64.49 cm; acceptable range [63.20, 65.78] cm.

Figure 11: **Example benchmark record for S3: Human–Product.** Task 2 entries include a product reference image, product-level metric dimensions, a human anchor, and a single product–human scale relation.

Field	Value
task_id	T3_0166
scenario	S4_Hard_Auto_Discovery
source_task_id	T2_0140
source_task_type	T2
prompt_type	hard_auto_discovery
image_path	generated/task2/Qwen_Image_Edit_2511_1024/T2_0140.png
objects_included	wine bottle; human head/face
gt_ratios	wine bottle_to_human head/face: target ratio 1.270; acceptable range [1.185, 1.364].
prompt	Check whether object size proportions in this image are unrealistic. Reference lengths — "wine bottle": approximately 30.48 cm characteristic, LxWxH 8.3 x 30.5 x 30.5 cm; human part: "human head/face" approximately 24.0 cm. If wrong relative to these references, rescale "wine bottle" only; if already plausible, keep the image unchanged. Preserve object identity, pose, viewpoint, composition, background, and lighting.
reference_image_path	ABO/images/original/52/52d1c3b3.jpg
source_size_score	4

Figure 12: **Example benchmark record for S4: Hard Auto-Discovery.** S4 entries provide an erroneous source image and scale references, but do not explicitly provide the editable object, resize direction, or scale factor.

Field	Value
task_id	T3_0013
scenario	S5_Precise_Scale_Instruction
source_task_id	T1_0321
source_task_type	T1
prompt_type	precise_scale_instruction
image_path	generated/task1/FLUX_2_Fal/T1_0321.png
objects_included	car; tennis racket
gt_ratios	car_to_tennis racket: target ratio 6.560; acceptable range [6.303, 6.827].
prompt	Shrink the tennis racket by an exact scale factor of 0.790000 while keeping the car completely unchanged. Strictly preserve the original identity, background, lighting, and composition, making absolutely no extra edits beyond this size correction.
edit_plan	edit target: tennis racket; reference object: car; scale factor: 0.790; direction: shrink.
pair_key	car_to_tennis racket
generated_ratio	5.200
target_ratio	6.560
reference_image_path	null
source_size_score	4

Figure 13: **Example benchmark record for S5: Precise Scale Instruction.** S5 uses the same type of erroneous source image as S4, but directly specifies the target object, reference object, resize direction, and numeric scale factor.

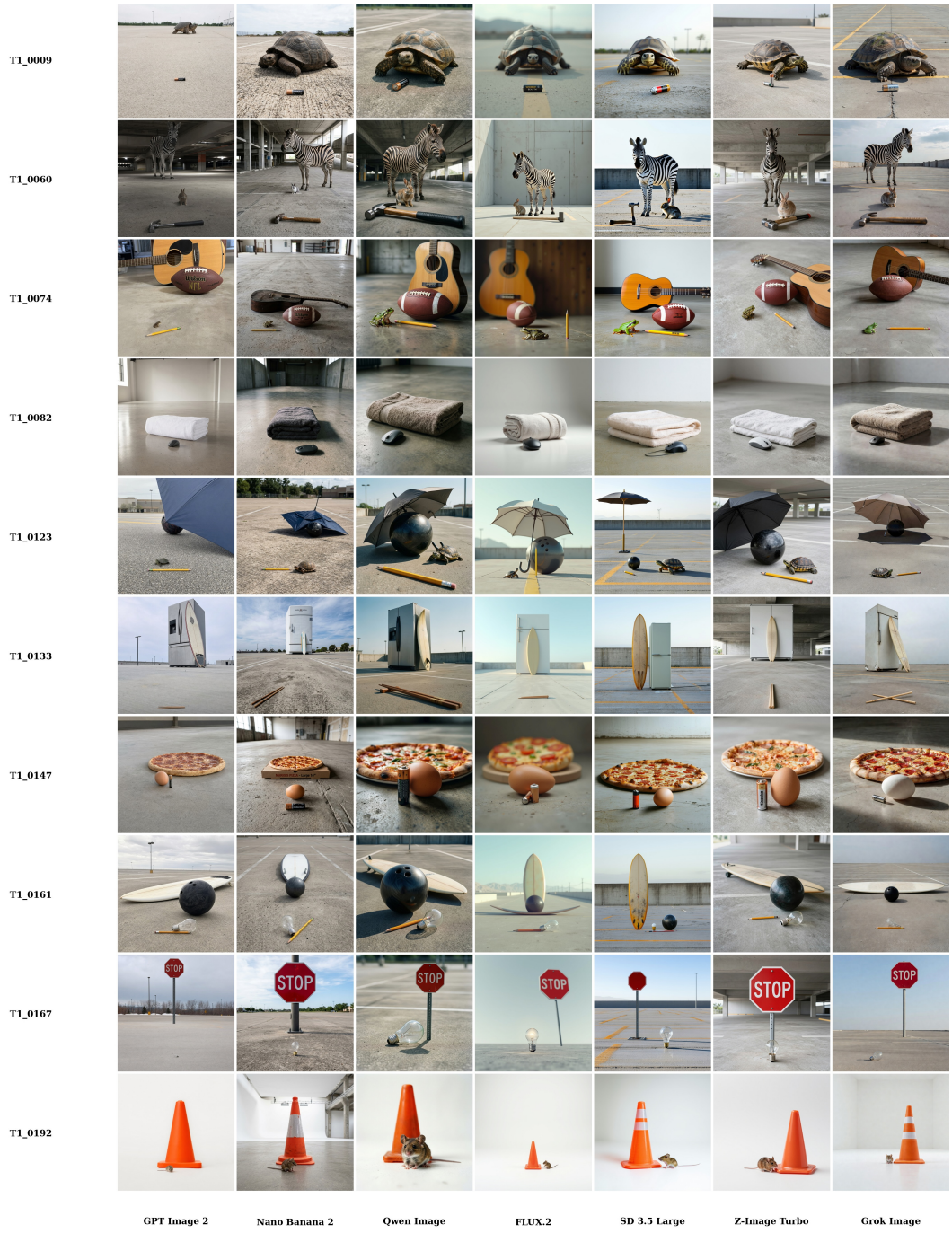

Figure 14: **Task 1 S1 Natural Depth examples.** Each row corresponds to one common-object prompt and each column corresponds to an evaluated generator. S1 allows perspective and depth ordering, so generated images may use near–far placement while still being evaluated for real-world relative scale.

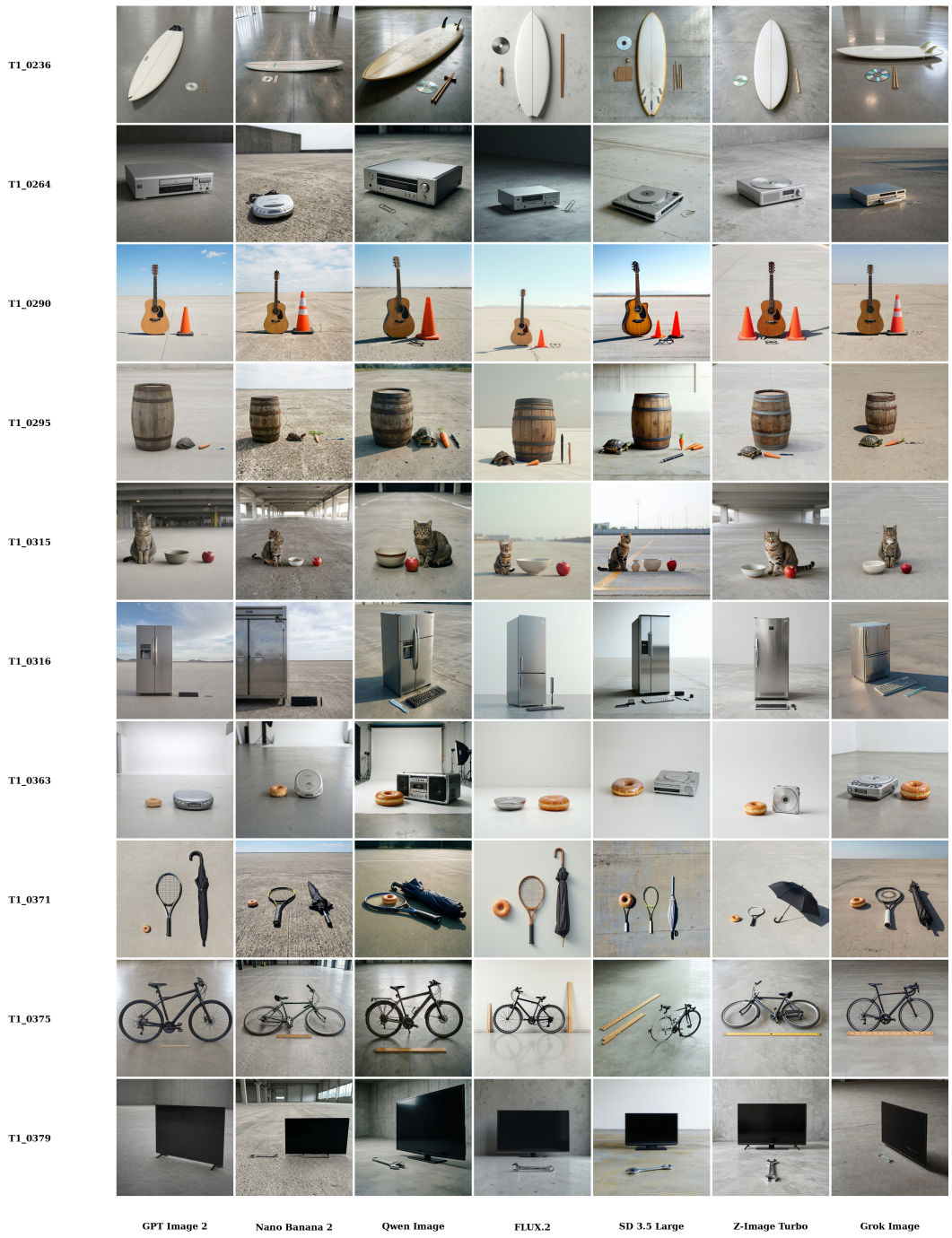

Figure 15: **Task 1 S2 Same-Plane examples.** Objects are prompted to lie on the same depth plane, reducing perspective ambiguity and making relative scale errors more directly visible in image space.

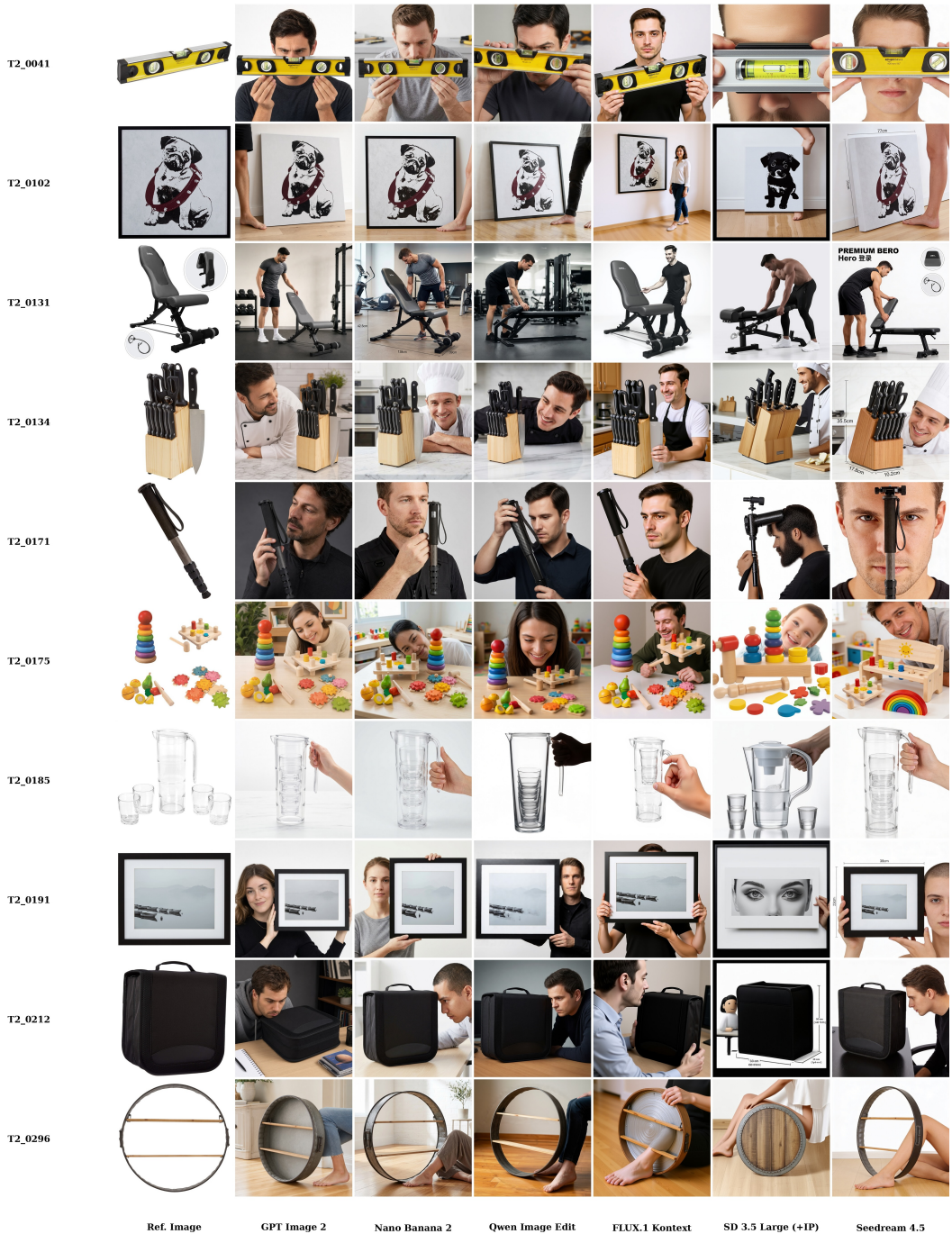

Figure 16: **Task 2 human-product generation examples.** Each row contains one product reference image and the corresponding model outputs. Prompts provide product dimensions and require a human anchor, so the task tests whether the model can translate explicit metric information into plausible human–product proportions.

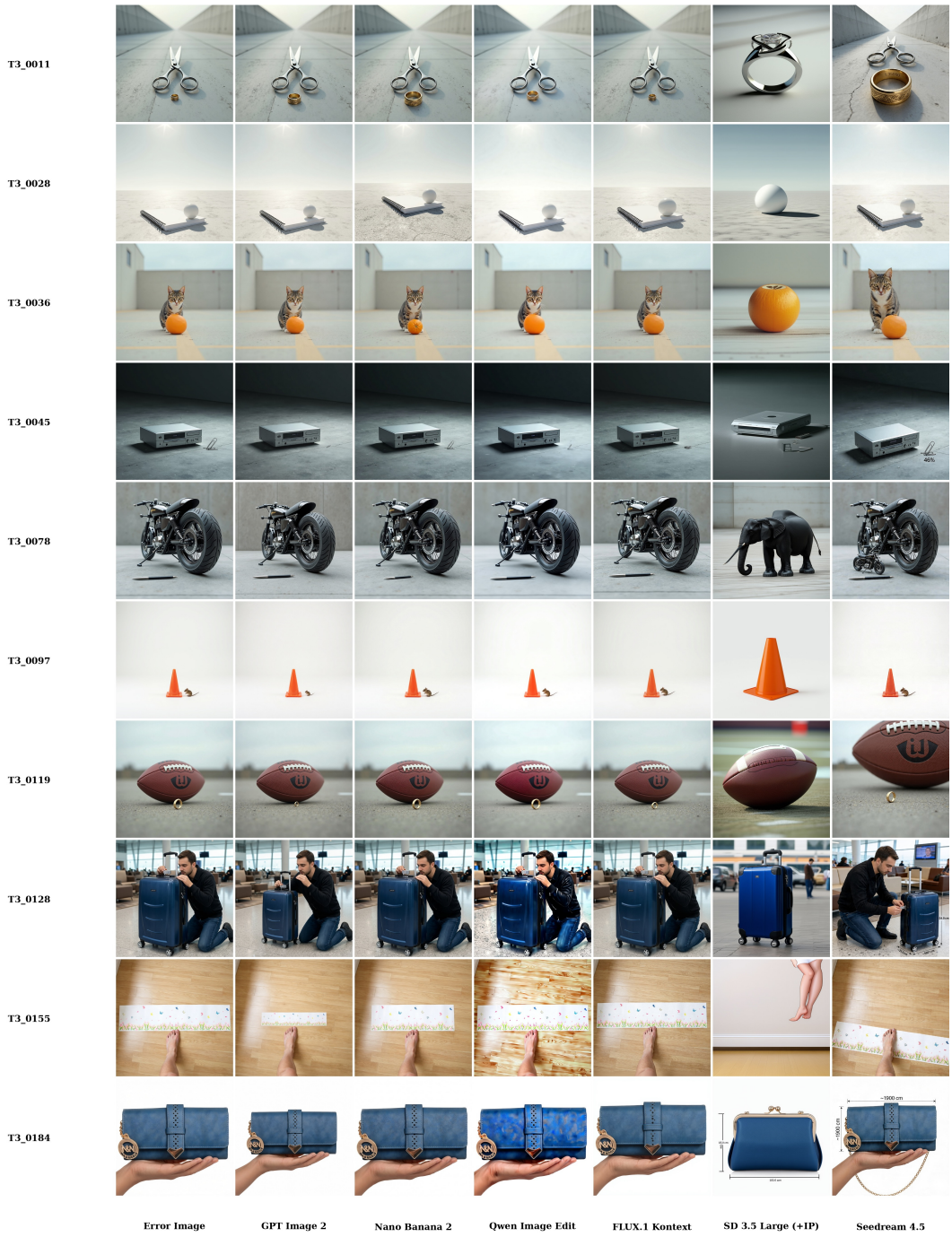

Figure 17: **Task 3 general-purpose editor correction examples.** The source image contains a known scale error from Task 1 or Task 2. General-purpose editors are asked to correct the error, either through auto-discovery or through a precise resize instruction depending on the scenario.

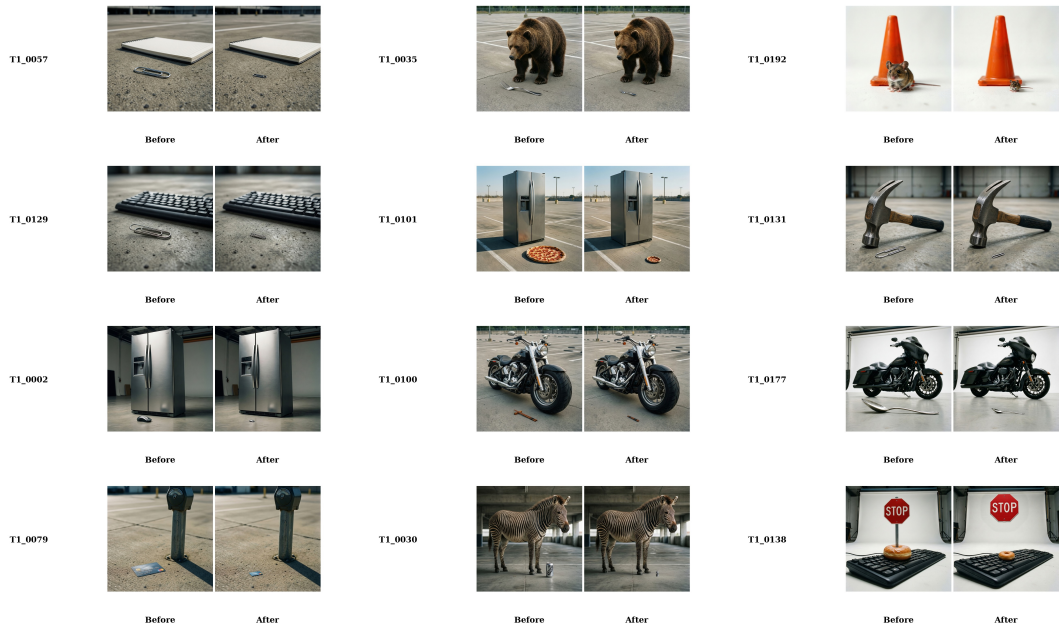

Figure 18: ***Rescale* corrections on S1 Natural Depth examples.** Each example shows the original generated image and the corrected output. *Rescale* uses the diagnosed scale relation to resize the local target while preserving the natural-depth composition.

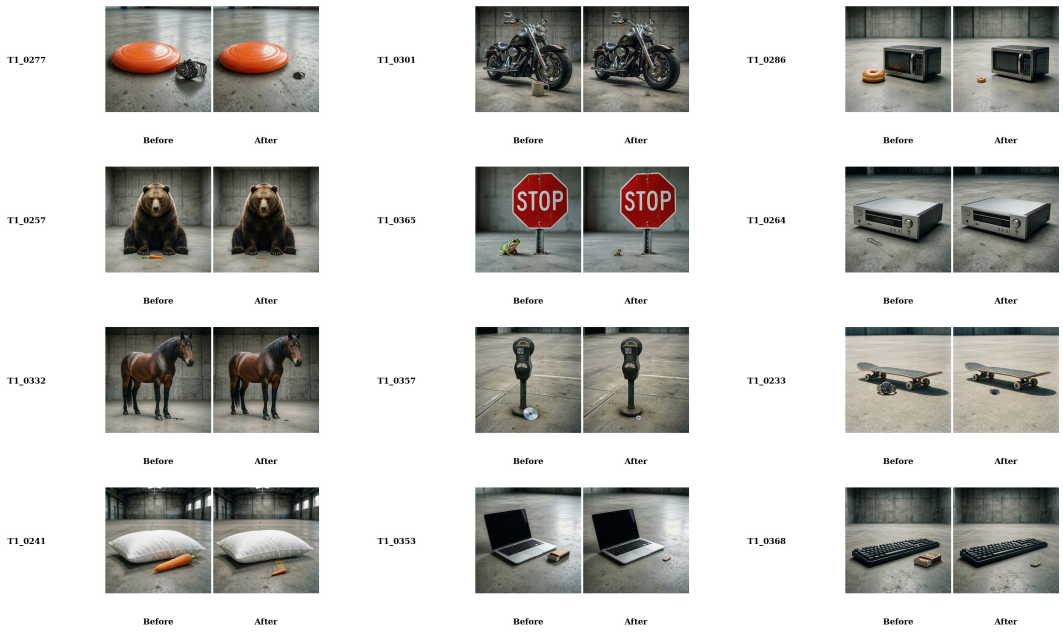

Figure 19: ***Rescale* corrections on S2 Same-Plane examples.** Because S2 constrains objects to the same depth plane, scale errors are visually direct; the correction mainly requires accurate local resizing and reinsertion.

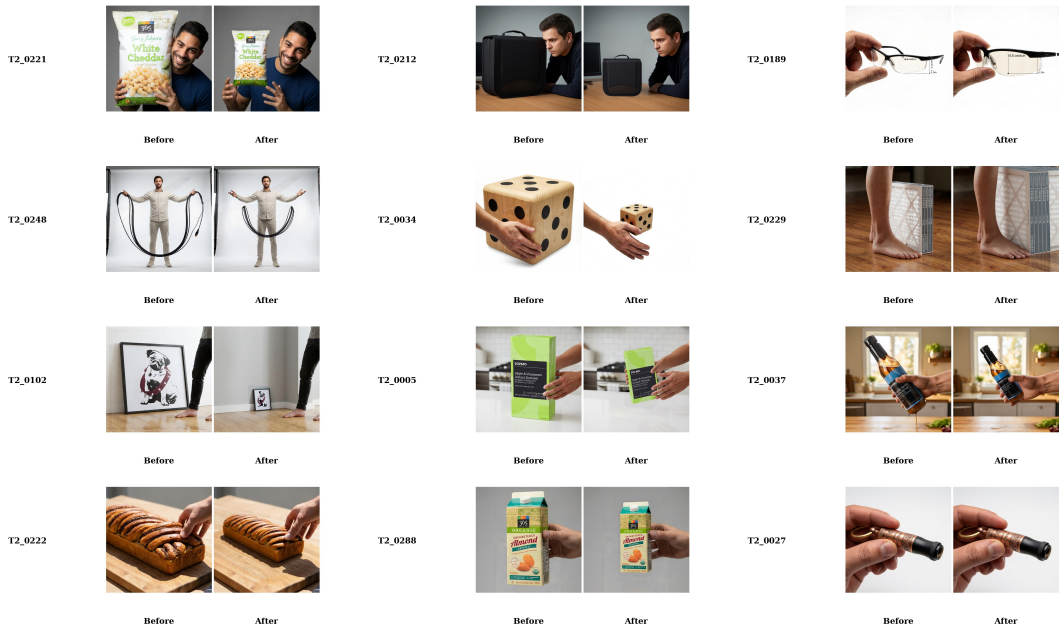

Figure 20: ***Rescale* corrections on S3 Human–Product examples.** These examples show corrections where the product is resized relative to a human anchor while preserving the product identity and the surrounding interaction context.

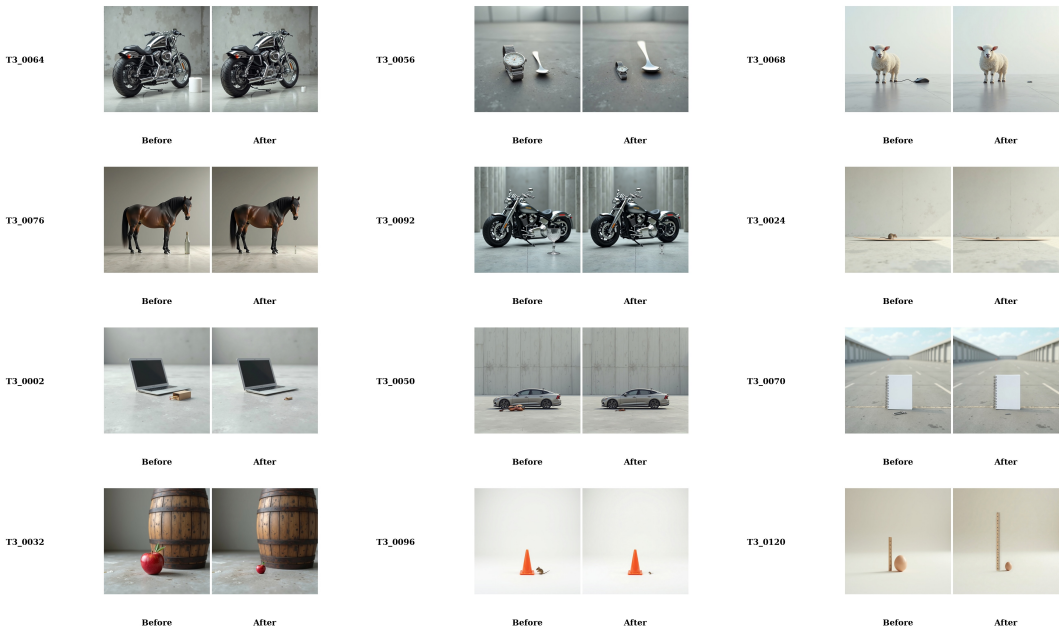

Figure 21: ***Rescale* corrections on S4 Hard Auto-Discovery examples.** In S4, the system must infer whether a scale error exists, identify the editable target, estimate the correction direction and factor, and then execute the local edit.

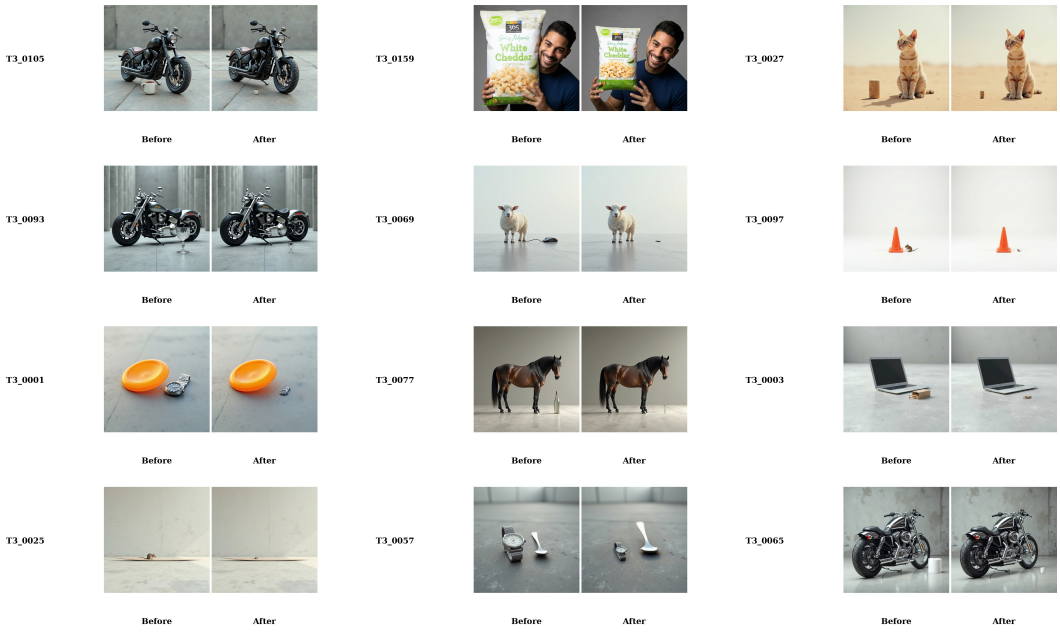

Figure 22: ***Rescale* corrections on S5 Precise Scale Instruction examples.** In S5, the target object, reference object, direction, and scale factor are given explicitly, isolating the localized resize-and-reinsert capability from scale-error diagnosis.

Figure 23: **Failure cases for S4 Hard Auto-Discovery.** Since the model must both diagnose and execute the correction, failures can occur at either stage: missing the scale anomaly, estimating an inaccurate factor, or applying a visually plausible but insufficient edit.

Figure 24: **Failure cases for S5 Precise Scale Instruction.** Even with the target and scale factor specified, local insertion can introduce artifacts, alter object identity, or fail to preserve the intended support region after resizing.